Estudo Orientado em Engenharia Informática 2022/2023

Introduction to the course

Topics in writing Technical Documents

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Parts of this presentation were originally prepared by Prof. Bruno Martins (IST)

Summary

- How to write technical documents
- Presenting results
- Some useful tools
- Handling data

- About this course

How to write technical documents

- Introduction
- How to organize your report / article
- Presenting results in technical reports
- Practical usage of tools

A thesis and its presentation

- **Definition for thesis:**
  - *a statement supported by rigorous and reasonable arguments*
- **Definition for scientific article (*or dissertation*):**
  - *document that presents and defends a thesis*
- **Definition for technical report:**
  - *document that presents technical/scientific contributions to the resolution of a problem (e.g., engineering details of a solution)*

General structure for a CS article/report describing original work

- Abstract
- **Introduction**
- **Background**
  - **Concepts**
  - **Related Work**

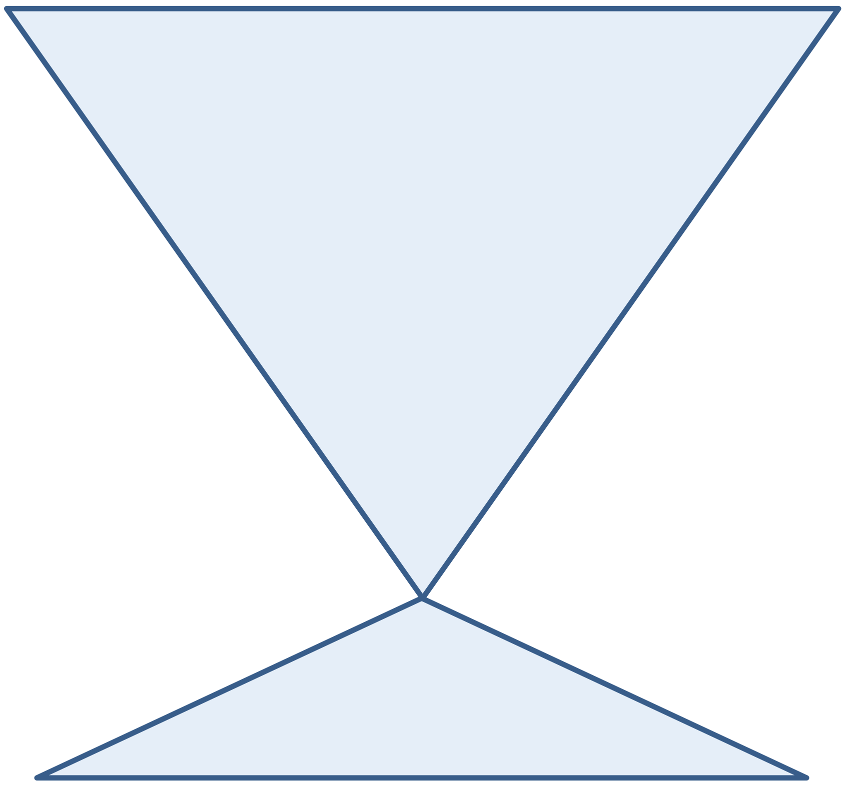
- **Data and Methods**
- **Results**
- **Discussion**

- **Conclusions**
  - **Conclusions**
  - **Future Work**
- Bibliography
- Supporting Materials

This can vary widely!

The Big Idea for Dissertation or Project

- Go from the General to the particular
  - Introduction
  - Related Work
  - Methods
  - …
  - Results
  - Discussion
  - Conclusion



Abstract and introduction

- Contextualize the problem being addressed
- Motivate the reader to the importance of adequately addressing the problem
- Present an hypothesis for a solution *(the thesis)*
- Overview the main results
- Present the organization of the report and Gantt Chart

Section giving the Background

- Present important concepts (*optional*)
  - Particularly important in cross-disciplinary work, that addresses questions from different sub-areas of CS
- Overview on the state of the art
  - Organize this section according to different ideas/themes, rather than by author or individual publication
  - Define all the related work
    - What others have been doing
    - How does your project fits in the overall organization
- Relevance to the problem being addressed

Section(s) detailing the contributions (Methods)

- Focus on the new ideas, i.e. on the work that was developed and that is being presented
- Describe clearly and systematically how the problem was addressed
  - one (sub-)section for each important contribution
  - ideally, one should be able to reproduce the work
- Should not include the "*dead ends*" encountered during the course of the work
  - Or should be clevery explained!

Sections presenting the results and discussion

- Results
  - Experimental (*empirical validation*)
    - present and describe the results of experiments
    - the results should demonstrate the important contributions
  - Design-oriented (*qualitative validation*)
    - matrix showing conformity with the requirements
    - studies with real users, presentation of case studies
- Discussion
  - Theoretical (*formal validation*)
    - demonstrations, comparisons against empirical data, ...
  - What should be expected, eventually subsequent testing in other settings
    - Confirmatory results

Section with the conclusions

- Present an overview of the contributions
  - organize from most to least important
- Present possible paths for future work
  - you, or other researchers in the same area, can later proceed on these research paths

Bibliographic references

- Related to the state of the art in the problem domain being addressed in the report
- Use up-to-date and authoritative references
  - use older references only to cite authorities in each area, or fundamental research in a domain
  - Pay attention to where the work was published
- All the works in the list of references should be cited in the body of the document

PRESENTING RESULTS IN TECHNICAL REPORTS

The presentation of results

- Aim for simplicity, correctness, and exactness
- Focus on the coherence of the data
- Always define your scope (limitations of current study)
- Support the discussion and motivate the reader

- The main objective purposes for presenting:
  - Describe the obtained (*experimental*) results – Don't just write the results. Do your results support your hypotesis?
  - Support exploration and the inference of conclusions
  - Tabulate and document what has been done
  - Compare different alternatives

Presentation of results

- Always qualify your results
  - Are your conclusions valid for ALL situations?
  - Clearly define boundaries of application
    - Should be defined in the introduction always but reminded in the Results, Discussion and conclusion sections
  - Identify perceivable uncertainties in your results

Tables vs. graphical charts

- Presentation in tables
  - better alternative to document the results
  - higher detail *(when there are few values)*
  - hard to establish patterns and general tendencies
- Presentation in charts / diagrams
  - use diagrams to communicate processes
  - better to communicate a specific message
  - harder to access specific values

Effective diagrams and charts

- In the diagrams, follow the “guidelines” of the area (*e.g., in CS, the standard is to use UML)*

- Avoid showing too much information (*keep it simple!*)
- Use coherent and sober elements, fonts, and colors
- Keep honest scales/axis in the charts
- Title should communicate the essential message
- Present enough data, so that the message is clear

- Identify the source of the data
- **Idea**: Place support data in an appendix (*or in the Web*)

Designing effective tables

- Use hierarchies in the table attributes
- Be consistent in the units for the presented values
- Justify numeric values to the left
- Clearly identify the rows and the columns:
  - avoid abbreviations, and preferably use complete names
- Show summary rows with the total values/averages
- Clearly identify the source of the data

Presentation using tables

Bad presentation

STATISTICS	SMALL	LARGE
Characters	18,621	1,231,109
Words	2,06	173,145
After stopping	1,2	98,234
Index size	1,31 Kb	109 Kb

Table 1: Collections used in the experiments

✓ Good presentation

	Collection	
	Small	Large
Size (Kb)	18,621	1,231,109
Index size (Kb)	1,310	109,000
Number of words	2,060	173,145
Words after stopping	1,200	98,234

Table 1: Collections used in the experiments

Source for the example : Writing for Computer Science, Justin Zobel

What about code?

- It's almost never a good idea to show up code
  - Even a "little bit" of SQL…
  - If **really necessary**, the idea should be discussed in the text
  - Typically, better to use a general algorithm

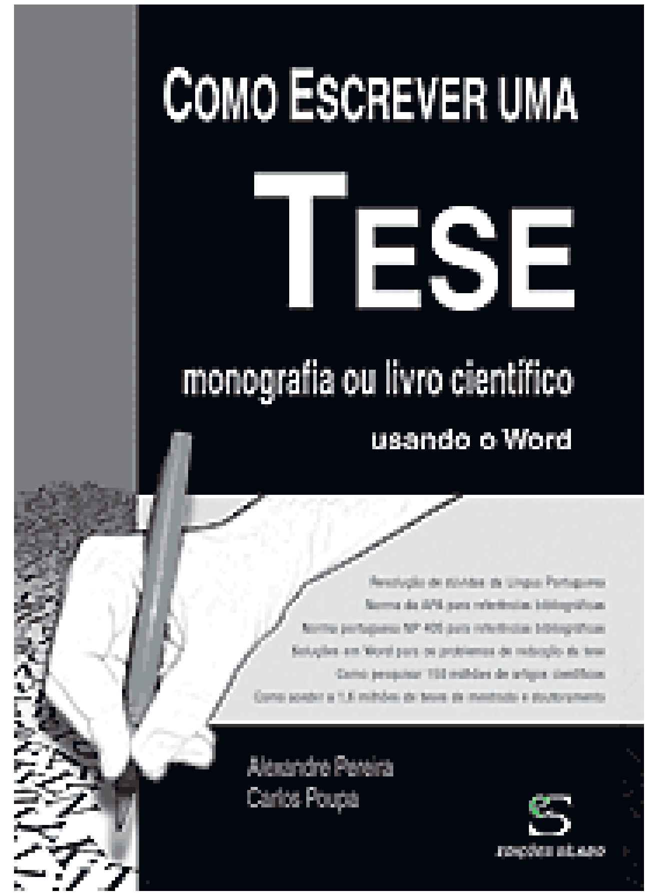
Algorithm 1 An algorithm with caption

Require: $n \geq 0$
Ensure: $y = x^n$
  $y \leftarrow 1$
  $X \leftarrow x$
  $N \leftarrow n$
  while $N \neq 0$ **do**
    if N is even **then**
      $X \leftarrow X \times X$
      $N \leftarrow \frac{N}{2}$ ▷ This is a comment
    else if N is odd **then**
      $y \leftarrow y \times X$
      $N \leftarrow N - 1$
    end if
  end while

PRACTICAL TOOLS

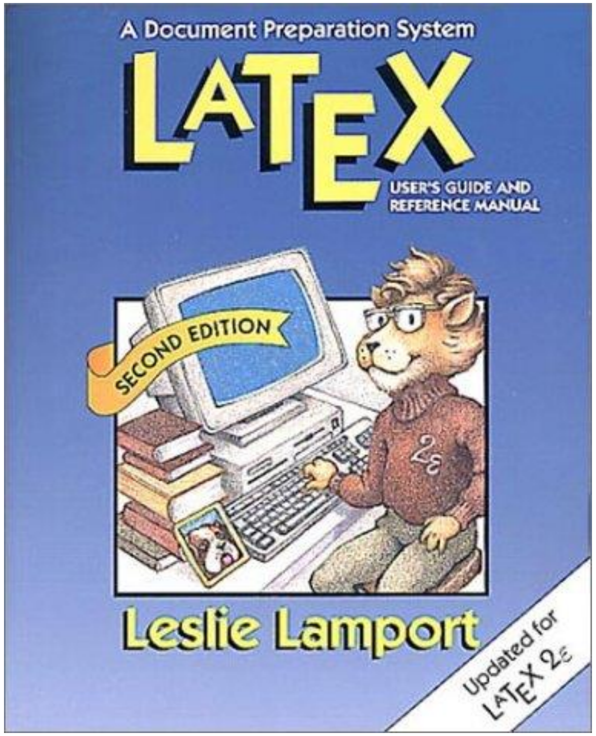
Microsoft Word

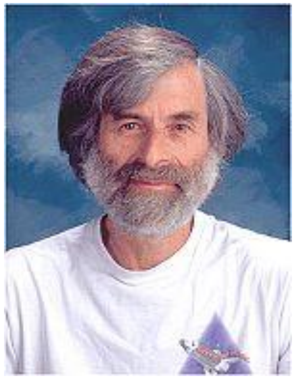
- In general, WYSIWYG tools are easy to use…
- There are books and other materials available, focusing on the writing of technical documents with Word…
- **Pay attention to the details!**

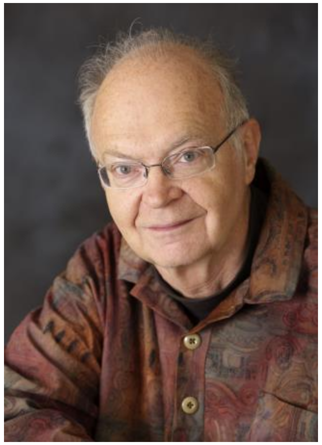


We argue for the use of LaTeX

- The LaTeX markup language and document preparation system, introduced by Leslie Lamport.
  - Based on TeX by Don Knuth







Why LaTeX

- With WYSIWYG tools (e.g., Word) the authors tend to focus on aesthetic/presentation issues...
  - The focus should be on the document structure/contents
- LaTeX forces authors to declare the **logical structure** of their documents...
  - LaTeX then chooses the appropriate *layout*
- LaTeX facilitates document ***portability***
  - many CS conferences and journals use *LaTeX templates* to translate logical structure into their specific *layout*

Compiling a LaTeX document

- Input can be manipulated with a text editor
- In an OS shell, one can:
  - compile: **latex myfile** or **pdflatex myfile**
  - Check the dvi file: **xdvi myfile**
  - convert dvi into PostScript: **dvips myfile**
  - convert PostScript into pdf: **ps2pdf myfile.ps**
  - check and print the PostScript: **gsview myfile**
- In Windows machines: MiKTeX (miktex.org)
- Online: overleaf.org [**recommended!**]
  - Shareable
  - Imports all packages.
  - Does all the strenuous work of managing dependencies and fixing references
  - Drawback: a project can only be shared to a limited number people

An Example LaTeX document

```
\documentclass[a4paper,11pt]{article}
\usepackage{latexsym}

\author{Bruno Martins}
\title{LaTeX example}
\begin{document}
\maketitle

\section{Start a section}
Here begins my article \ldots
\subsection{A subsection}
This is a subsection \ldots
\section{End}

\end{document}
```

LaTeX example

Bruno Martins

1. Start a section

Here begins my article…

1.1 A subsection

This is a subsection…

Different types of documents

- **Article** – has sections, subsections, …
- **Report** – has chapters, sections, …
- **Slides** – there are appropriate styles for creating presentations (see LaTeX package *beamer*)

- Several options for each document type:
  - Size of the textual font
  - Size of the paper sheets and *layout*
  - Number of columns in the document
  - ...

Typesetting mathematical expressions

- The **ease in typesetting math expressions**, equations, and symbols is one of the main advantages in using LaTeX:

```
To find the square of
the hypotenuse, add a
squared to b squared to
find c squared,

\begin{displaymath}
a^2 + b^2 = c^2
\end{displaymath}
```

$$a^2 + b^2 = c^2$$

```
\ldots it is clear that

\begin{equation}
\alpha > 0
\label{eq:eps}
\end{equation}

From Equation~\ref{eq:eps}
it follows that \ldots
```

$$\alpha > 0$$

Including graphics in LaTeX

- LaTeX has a package for including images in the PostScript format.
- The header of the LaTeX document should include the package:
  - **\usepackage{graphicx}**
- A figure can now be introduced in the document

```
\begin{figure}[ht]
\begin{center}
\includegraphics[width=140mm]{mypic.ps}
\end{center}
\caption{An example of a figure.}
\label{fig:example}
\end{figure}
```

Lists and enumerated lists

- There are 3 types of lists in LaTeX
  - Enumerated lists
  - Lists of items (one *bullet* per item)
  - Descriptions (each item has a descriptor)

```
\begin{enumerate}
\item Birds
\item Trees
\end{enumerate}
```

1. Birds
2. Trees

Tables in LaTeX

- There is a specific environment for creating tables

```
\begin{table}
\center
\caption{A Simple Table}
\label{tb:simple}
\begin{tabular}{r|rc}
C1 & C2 & C3 \\
\hline
A & B & C\\
D & E & F\\
\end{tabular}
\end{table}
```

Table 1: A Simple Table

C1	C2	C3
A	B	C
D	E	F

Bibliography and references

- Citations to bibliographic references can be made in the text using the command **\cite{}**:
  - details for each bibliographic reference given in BibTeX file
  - separate file controls the presentation style associated to the bibliographic references
- References to specific parts of a document (e.g., to specific sections, tables, figures, etc.) can be made in the text with the command **\ref{}**:

```
\section{Exemplo de Secção} \label{sec:ex}
...
So as I was saying in~\ref{sec:ex}
```

BibTeX documents

- BibTeX is a reference management tool for formatting lists of references, integrated with LaTeX.
- Different types of BibTeX entries, for different types of documents that can be referenced

```
@article{Dugan:2006:WCS:1127442.1127468,
  author = {Robert F. Dugan  and Virginia G. Polanski},
  title = {Writing for computer science: a taxonomy of writing tasks and general advice},
  journal = {Journal of Computing Sciences in Colleges},
  volume = {21},
  number = {6},
  year = {2006},
  issn = {1937-4771},
  pages = {191--203},
  numpages = {13},
  publisher = {Consortium for Computing Sciences in Colleges},
}
```

The *natbib* package

- The extension package *natbib* re-implements the **\cite{}** command, to work with both author-year and numerical citations.

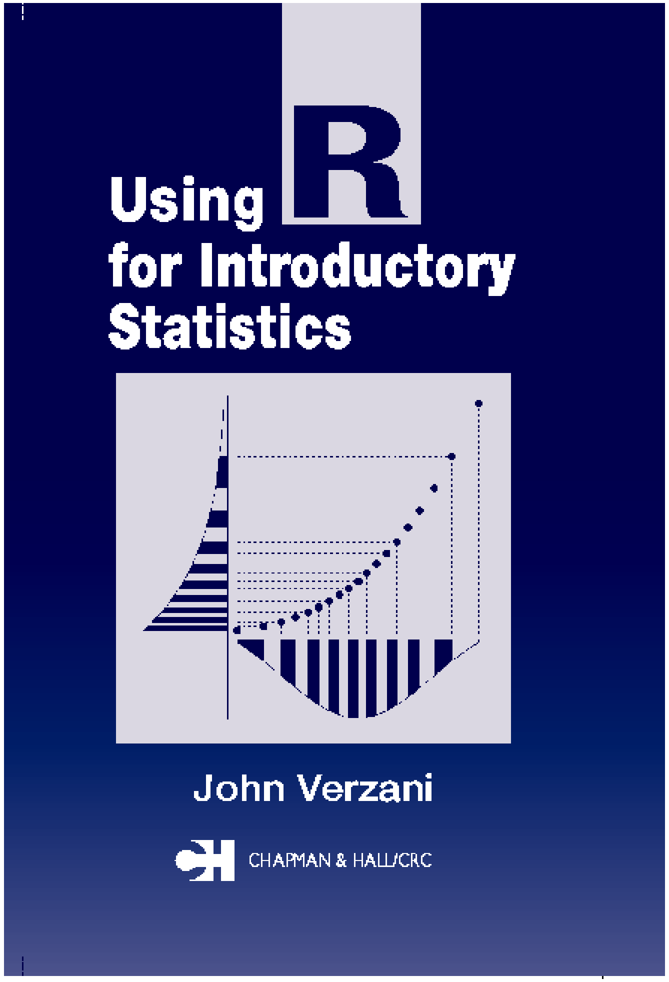
  - \citet{jon90} Jones et al. (1990)
  - \citet[chap. 2]{jon90} Jones et al. (1990, chap. 2)
  - \citep{jon90} (Jones et al., 1990)
  - \citep[chap. 2]{jon90} (Jones et al., 1990, chap. 2)
  - \citep[see][]{jon90} (see Jones et al., 1990)
  - \citep[see][chap. 2]{jon90} (see Jones et al., 1990, chap. 2)
  - \citet*{jon90} Jones, Baker, and Williams (1990)
  - \citep*{jon90} (Jones, Baker, and Williams, 1990)

Charts and Diagrams

- Dia (*Diagram Editor*)
- MS Visio
- MS Powerpoint
- Online: draw.io

- GNU plot
- R
- Matplotlib and seaborn (Python)
- Tikz package for LaTeX
- Microsoft Excel
- …

- Tables in LaTeX or in Microsoft Word



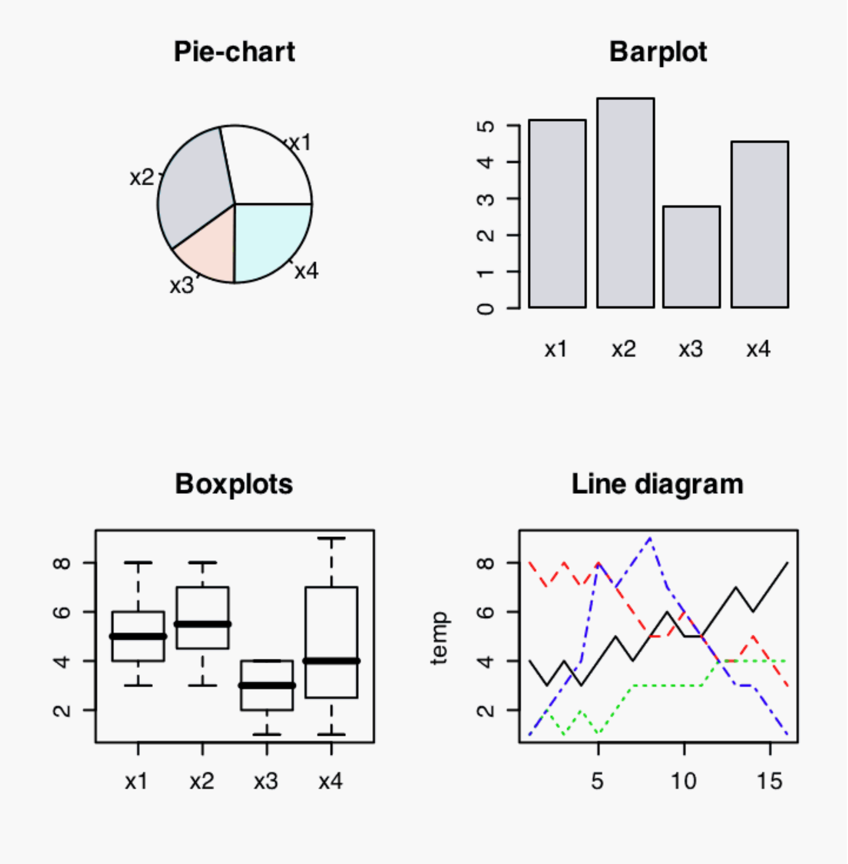
Example using R

```
x1 <- c(4, 3, 4, 3, 4, 5, 4, 5, 6, 5, 5, 6, 7,
    6, 7, 8 )
x2 <- c(8, 7, 8, 7, 8, 7, 6, 5, 5, 6, 5, 4, 4,
    5, 4, 3 )
x3 <- c(1, 2, 1, 2, 1, 2, 3, 3, 3, 3, 3, 4, 4,
    4, 4, 4 )
x4 <- c(1, 2, 3, 4, 8, 7, 8, 9, 7, 6, 5, 4, 3,
    3, 2, 1 )
df <- data.frame(x1, x2, x3, x4)

par(mfrow=c(2, 2))


pie(mean(df),main="Pie-chart")
barplot(mean(df), main="Barplot")
boxplot(df, main="Boxplots")
matplot(df,type="l", main="Line diagram")

par(mfrow=c(1, 1))
```



Source for the example: http://www.rensenieuwenhuis.nl/archive/r-sessions-14-multiple-graphs/

Last but not least…

- Don’t forget the **spell checker**
  - Your favorite text editor must have it enabled
  - E.g., *aspell* on Linux
- Or even a **style checker**
  - https://www.cs.umd.edu/~nspring/software/style-check-readme.html
  - https://github.com/vvcogo/style-check-html
- Or both: https://app.grammarly.com/

This course

About this course

- Purpose: Help in the organization of final report (MSc Thesis); thinking rigorously in Computer Science and making good presentations
- Grading
  - A written report according to a provided LaTeX template
  - A presentation before a jury consisting of the supervisor and a PhD examiner (DI or external professor)
- The report and presentation slides must be delivered before the beginning of the 1st Semester exams
- The presentation must be made before end date of the 1st semester exams

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Bibliographic research

Andre Falcao (aofalcao@fc.ul.pt)

Parts of this presentation were originally prepared by Prof. Bruno Martins (IST) and Alysson Bessani (FCUL)

Motivation

- Typically, students do not know how to present information
- Here follows a very poor example of a written text

Rotational mechanics is one of the most important factors for the quality of life of the urban man. Among the various problems still unresolved we have orthorhombic kinetics and the definition of interfaces for home users. The use of Deepest Learning techniques is the best strategy to solve these problems. Doctor John Smith gives a good overview of the topic [86]

Comment [A5]: Who said so?

Comment [A6]: Who said so?

Comment [A7]: Personal appreciation and no citation

Comment [A8]: Very bad way of framing a complex and multiple citation Makes a personal and qualitative affirmation about a work and cites in a absurd way the work of Smith

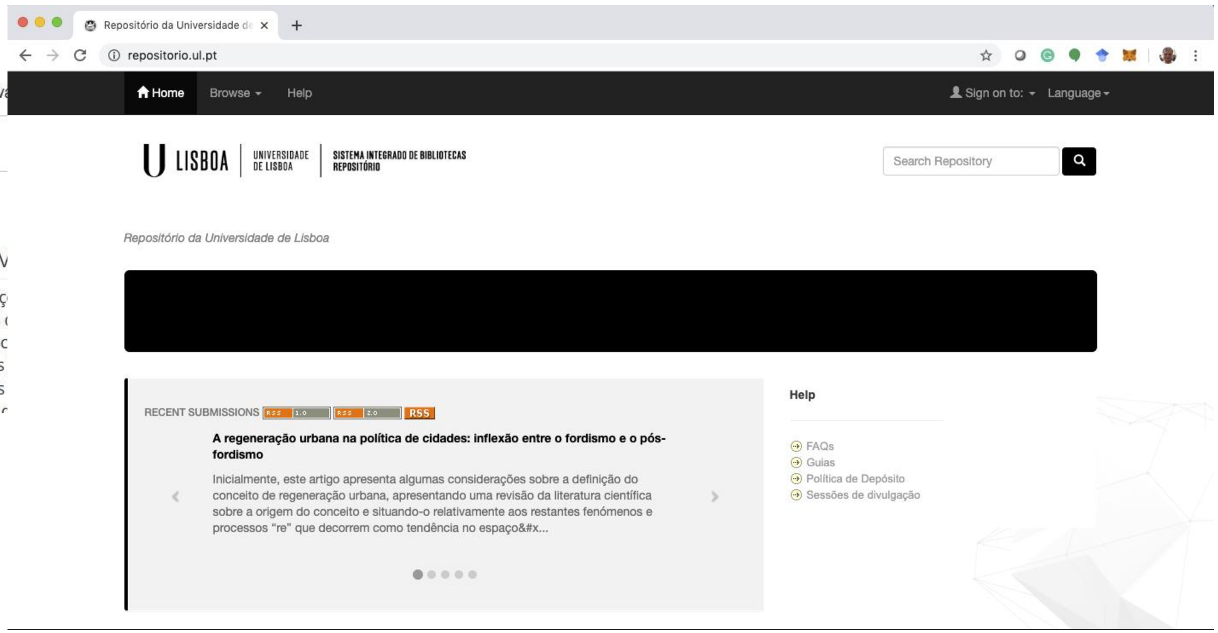
Bibliographic research

- New discoveries are published on journal articles, conference proceedings, books, thesis, etc.
- Bibliographic research supported on computers:
  - All-purpose search engines can be used for academic purposes, and some search engine companies also have specialized tools
  - Bibliographic databases giving information about finding books and articles
  - Digital libraries focused on Comp. Science materials
- Tools have advanced retrieval mechanisms, combine searching and browsing

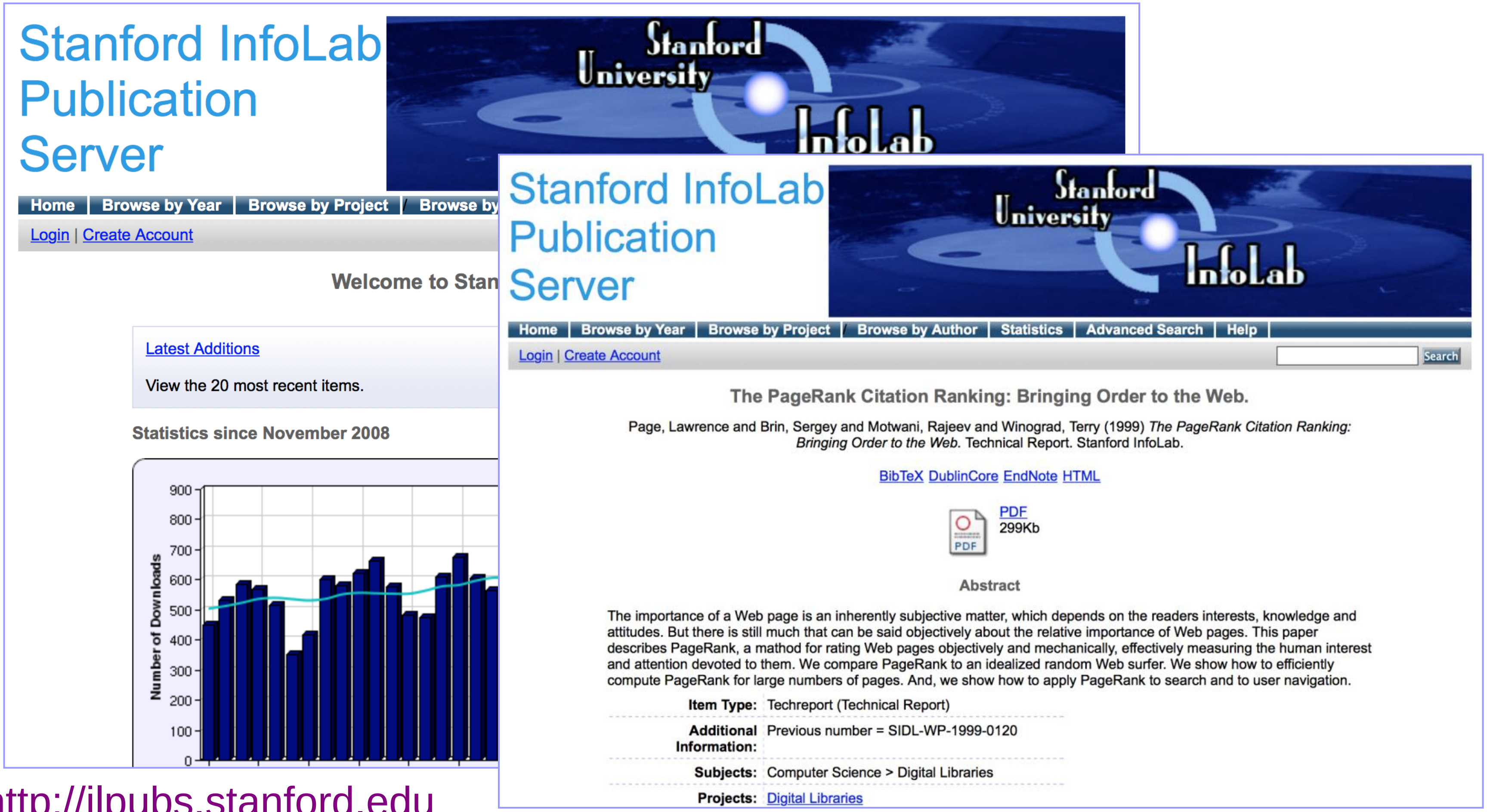
FCUL and ULisboa

https://catalogo-bibliotecas.ulisboa.pt/

http://repositorio.ul.pt



Publication servers at different institutions (e.g., based on e-prints)

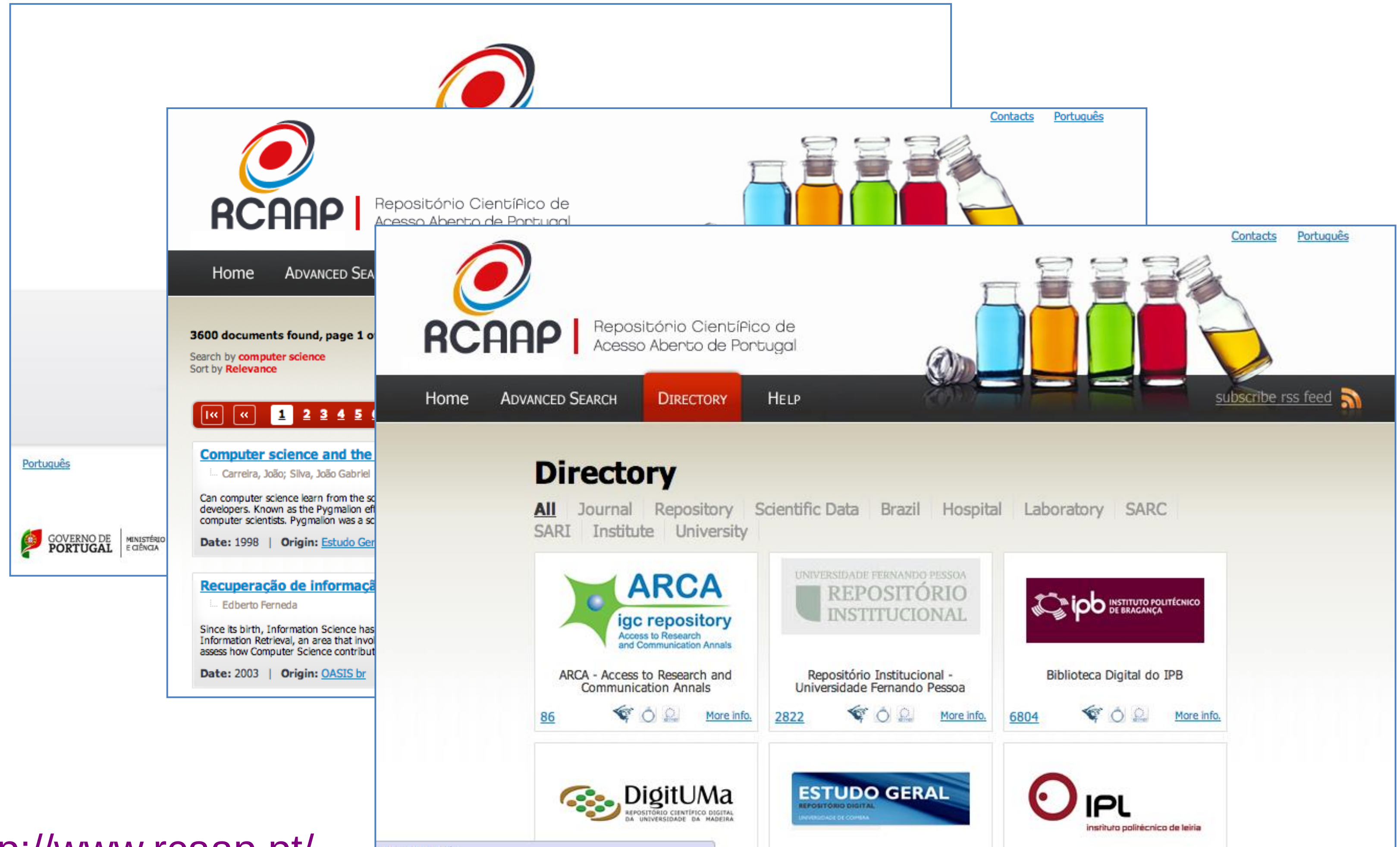


http://ilpubs.stanford.edu

What about course materials?

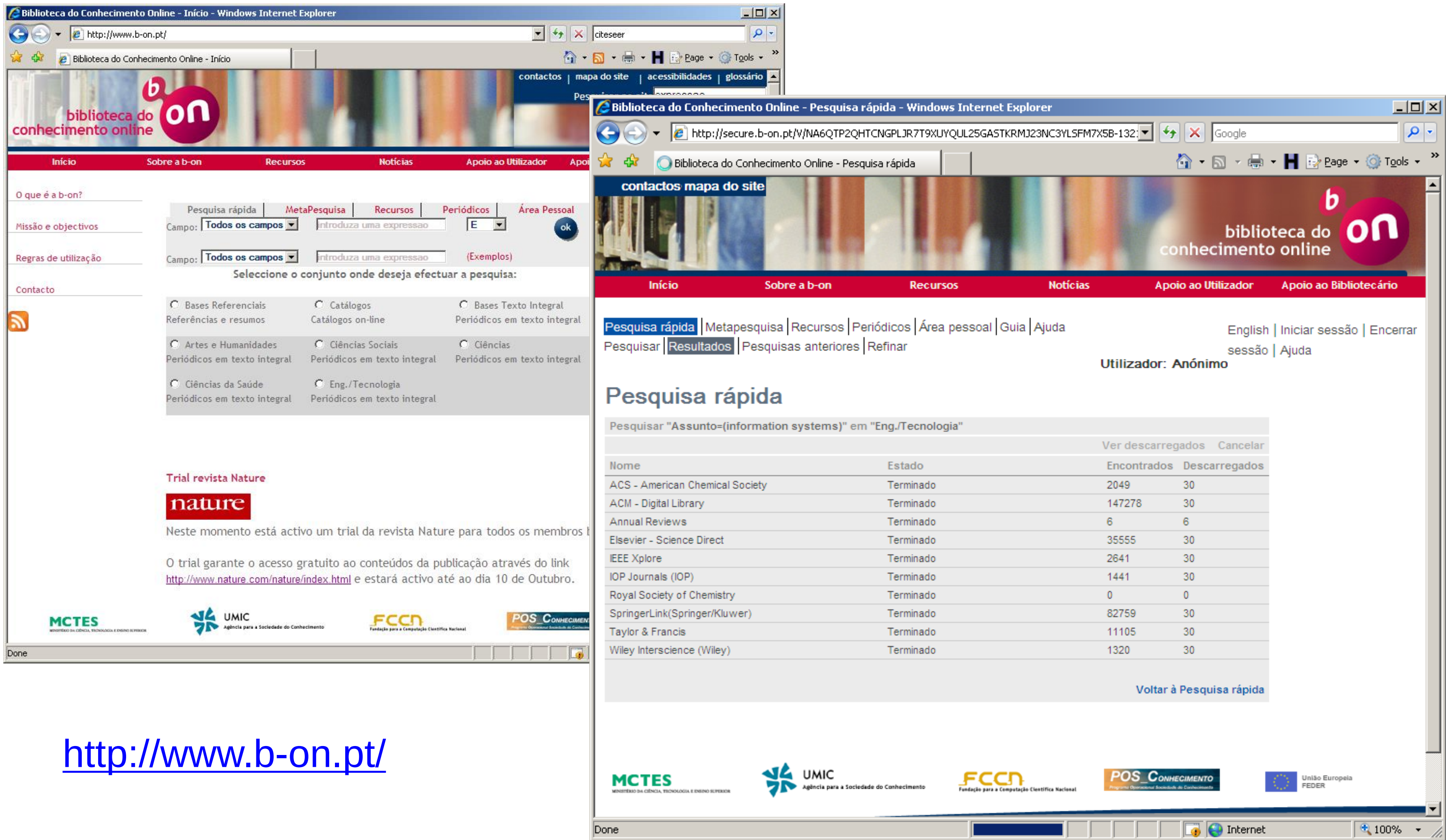
- Avoid quoting your course lecturer directly!
  - E.g.: Eustáquio, T. Slides and lecture notes from the course "Optimização Concorrencial" FCUL, Lisboa
  - Not wrong in itself but "amateurish"
- That's Ok to use course textbooks
  - Check for the latest edition!
- Always good to use landmark papers pointed by the lecturer

RCAAP



http://www.rcaap.pt/

b-on

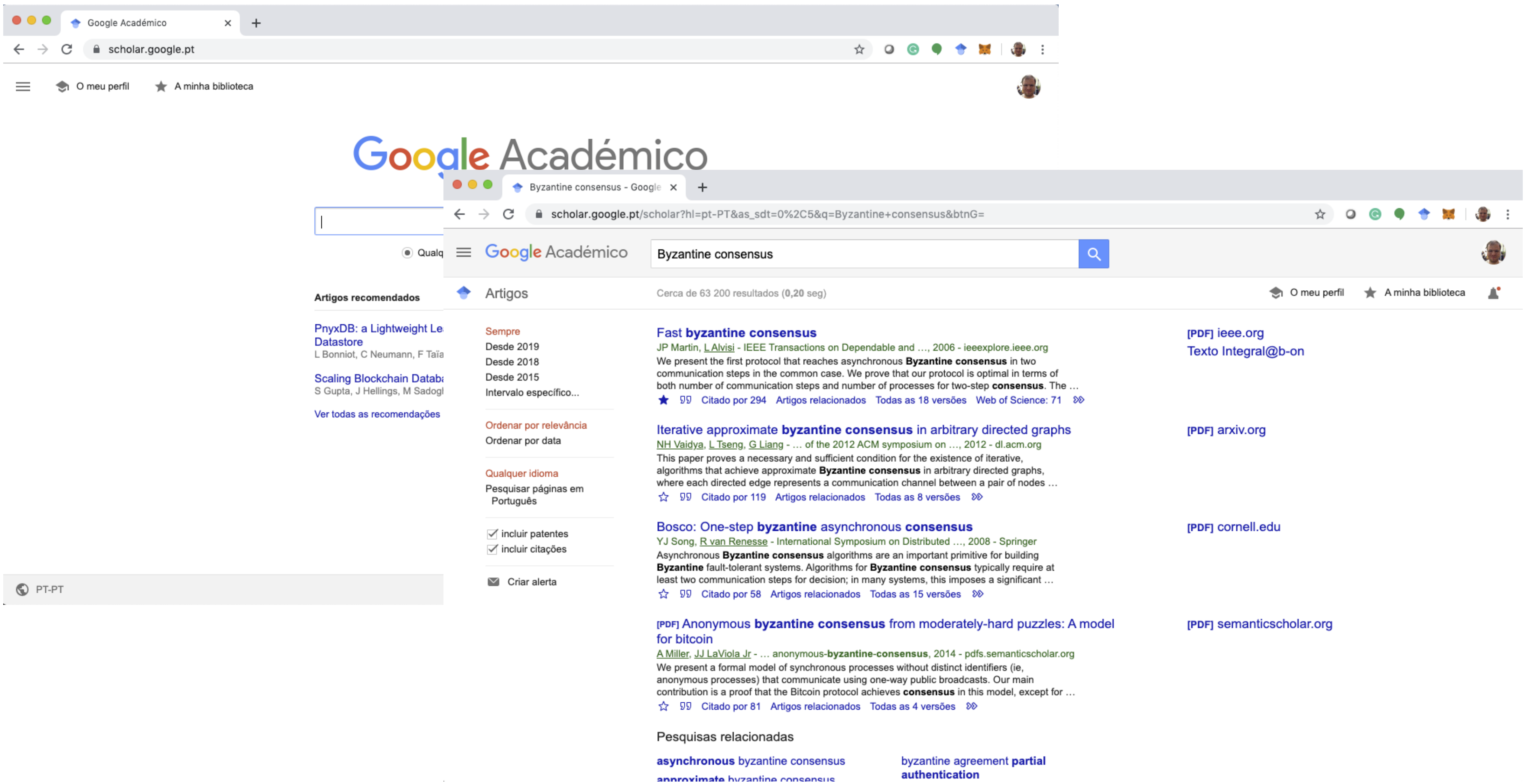


http://www.b-on.pt/

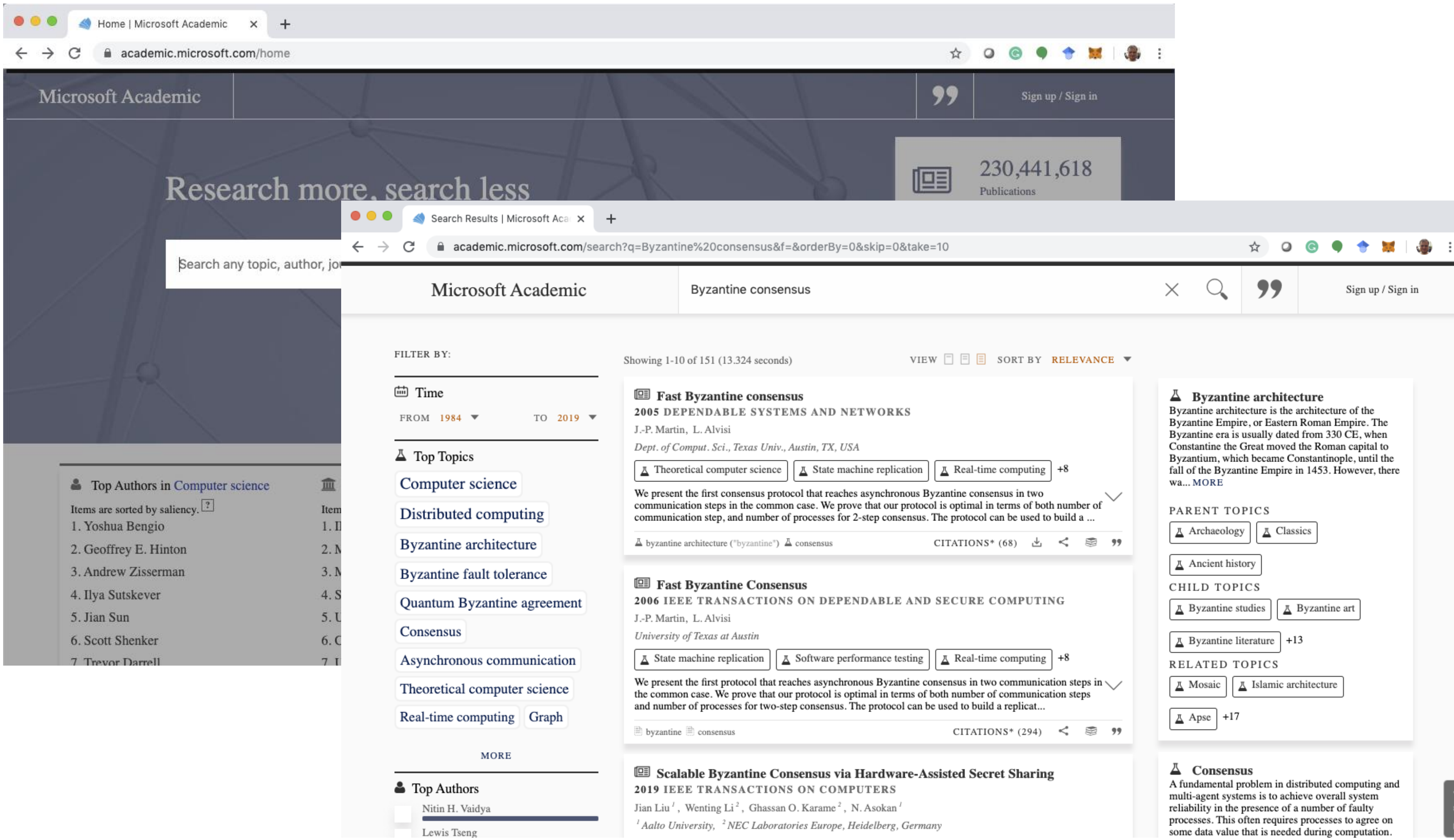
arXiv e-print service and the Computing Research Repository (CoRR)

http://arxiv.org

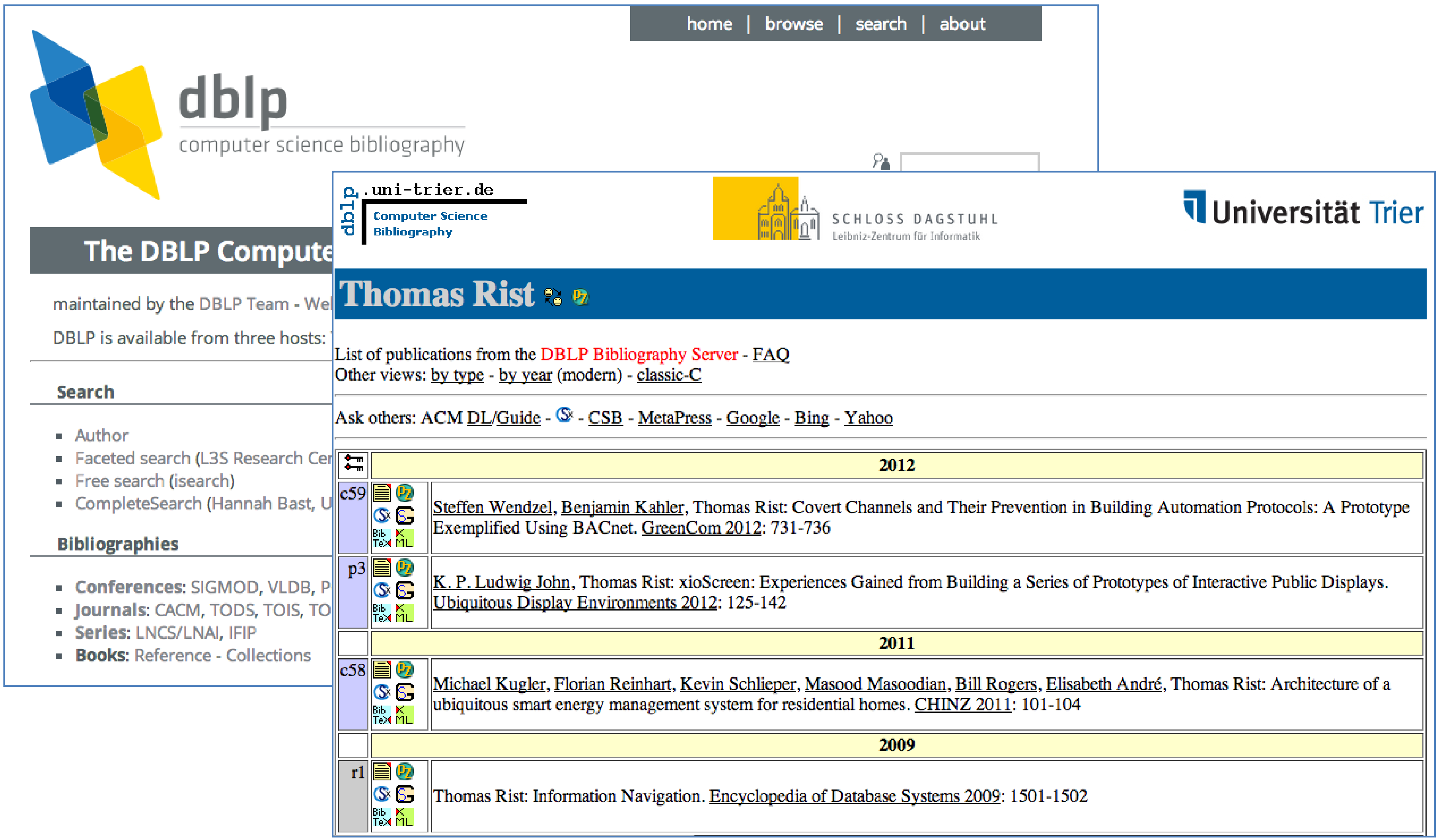
Google Scholar



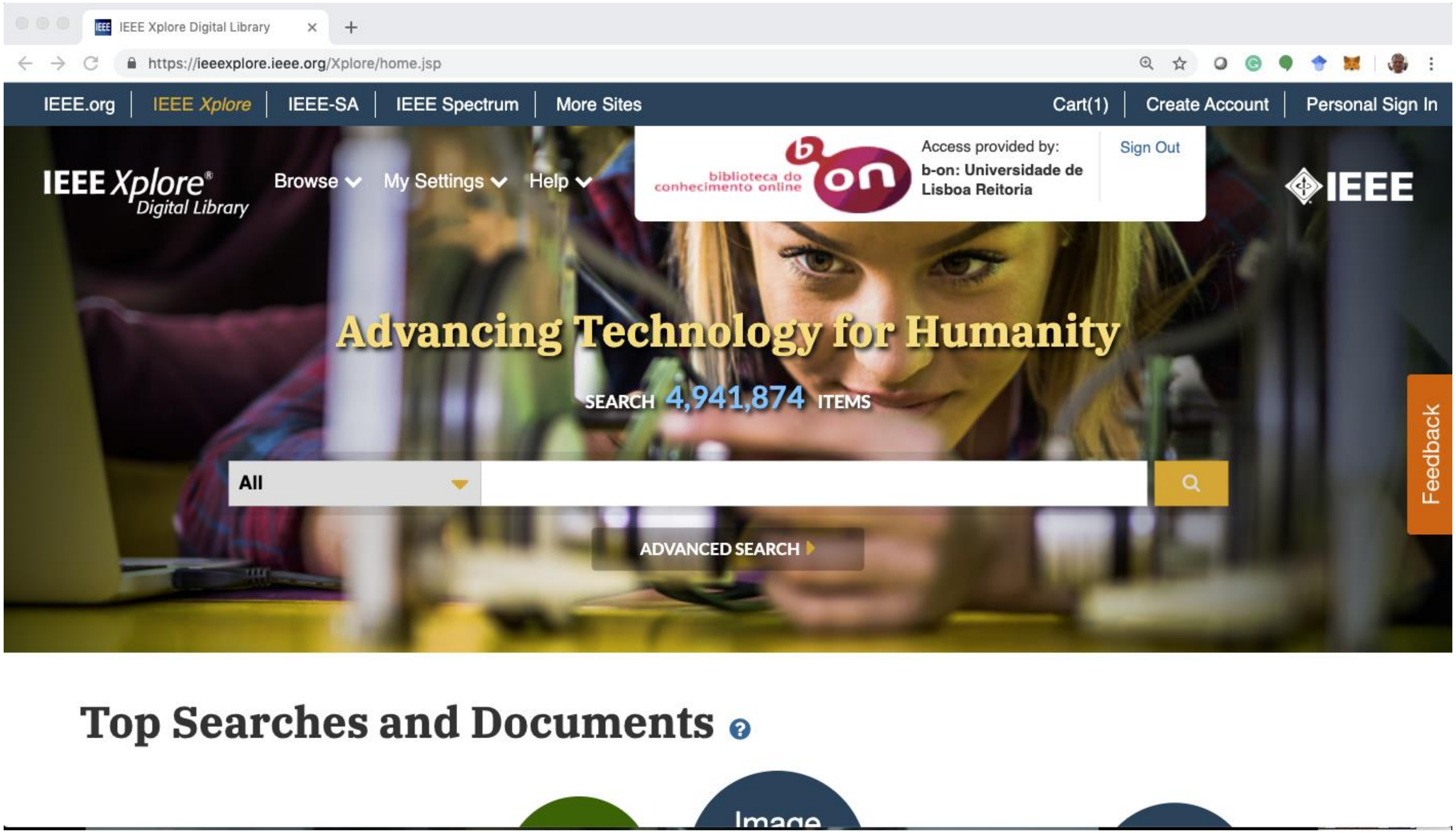
Microsoft Academic



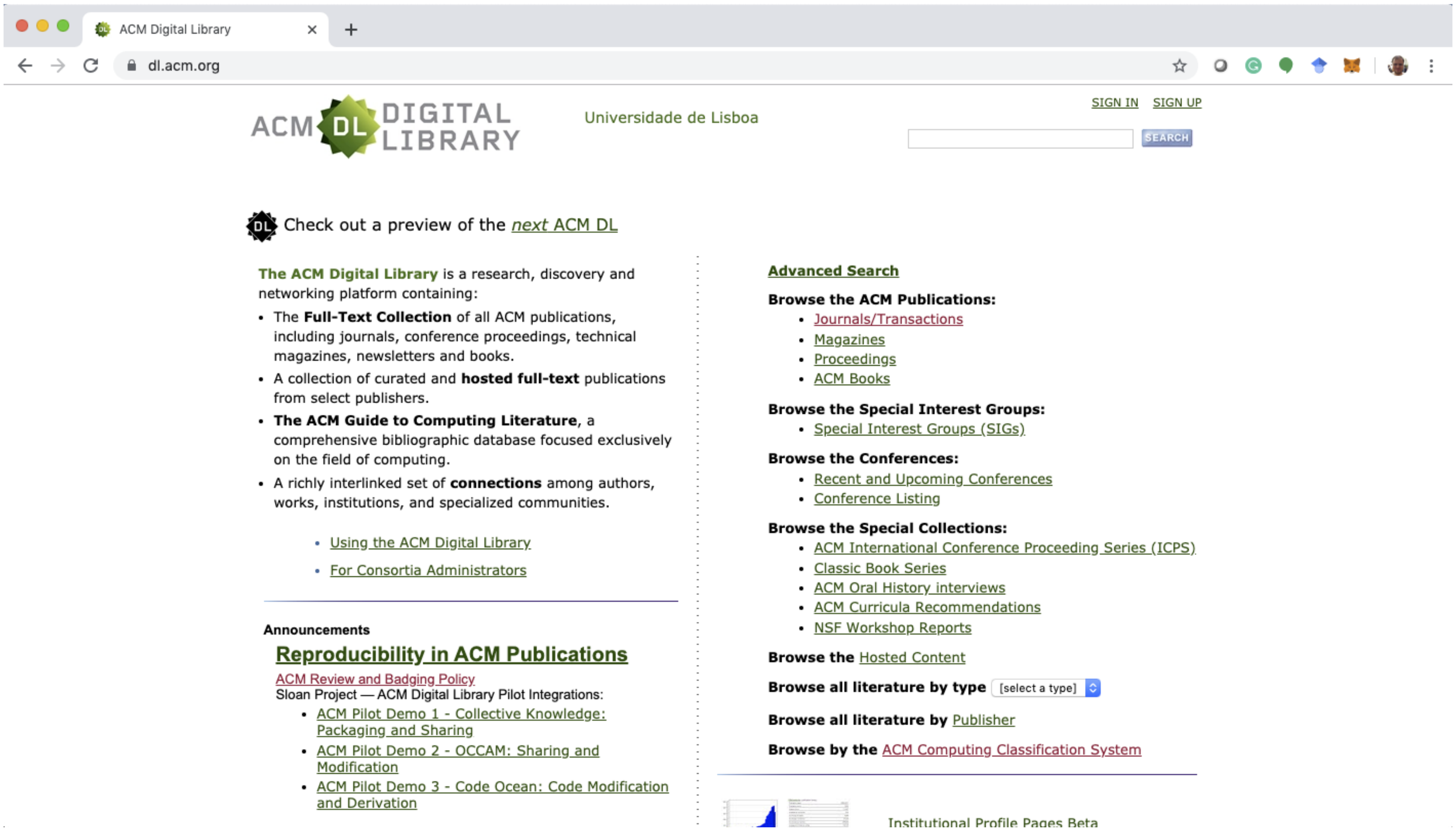
DBLP


home | browse | search | about
dblp
computer science bibliography
The DBLP Compute
maintained by the DBLP Team - We
DBLP is available from three hosts:
Search
Author
Faceted search (L3S Research Cer
Free search (isearch)
CompleteSearch (Hannah Bast, U
Bibliographies
Conferences: SIGMOD, VLDB, P
Journals: CACM, TODS, TOIS, TO
Series: LNCS/LNAI, IFIP
Books: Reference - Collections
dblp.uni-trier.de
Computer Science Bibliography
SCHLOSS DAGSTUHL
Leibniz-Zentrum für Informatik
Universität Trier
Thomas Rist
List of publications from the DBLP Bibliography Server - FAQ
Other views: by type - by year (modern) - classic-C
Ask others: ACM DL/Guide - CSB - MetaPress - Google - Bing - Yahoo
2012
c59
Steffen Wendzel, Benjamin Kahler, Thomas Rist: Covert Channels and Their Prevention in Building Automation Protocols: A Prototype Exemplified Using BACnet. GreenCom 2012: 731-736
p3
K. P. Ludwig John, Thomas Rist: xioScreen: Experiences Gained from Building a Series of Prototypes of Interactive Public Displays. Ubiquitous Display Environments 2012: 125-142
2011
c58
Michael Kugler, Florian Reinhart, Kevin Schlieper, Masood Masoodian, Bill Rogers, Elisabeth André, Thomas Rist: Architecture of a ubiquitous smart energy management system for residential homes. CHINZ 2011: 101-104
2009
r1
Thomas Rist: Information Navigation. Encyclopedia of Database Systems 2009: 1501-1502

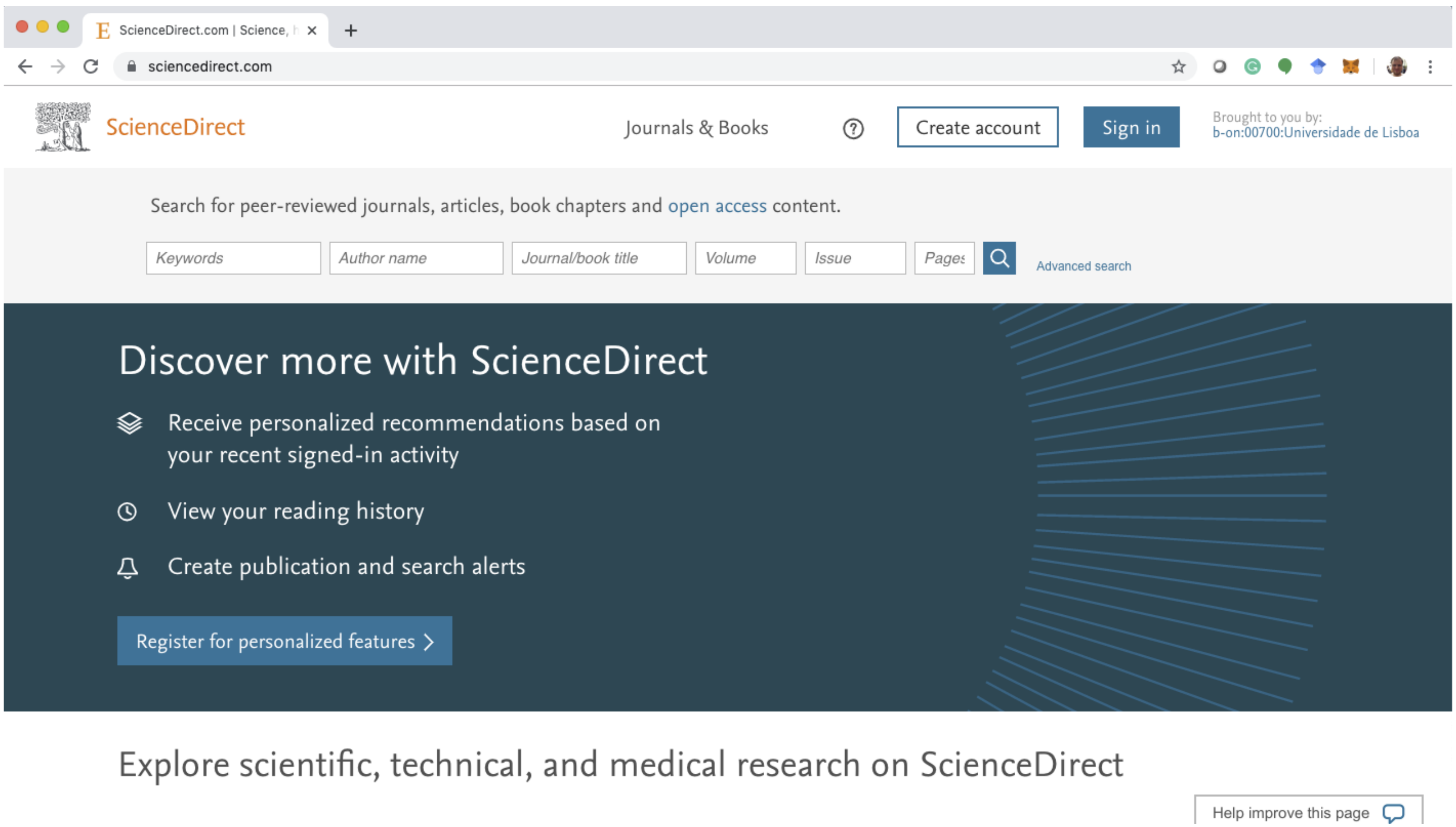
IEEE Xplore


IEEE Xplore Digital Library
https://ieeexplore.ieee.org/Xplore/home.jsp
IEEE.org
IEEE Xplore
IEEE-SA
IEEE Spectrum
More Sites
Cart(1)
Create Account
Personal Sign In
IEEE Xplore® Digital Library
Browse
My Settings
Help
biblioteca do conhecimento online
b-on
Access provided by:
b-on: Universidade de Lisboa Reitoria
Sign Out
IEEE
Advancing Technology for Humanity
SEARCH 4,941,874 ITEMS
All
ADVANCED SEARCH
Feedback
Top Searches and Documents
Image

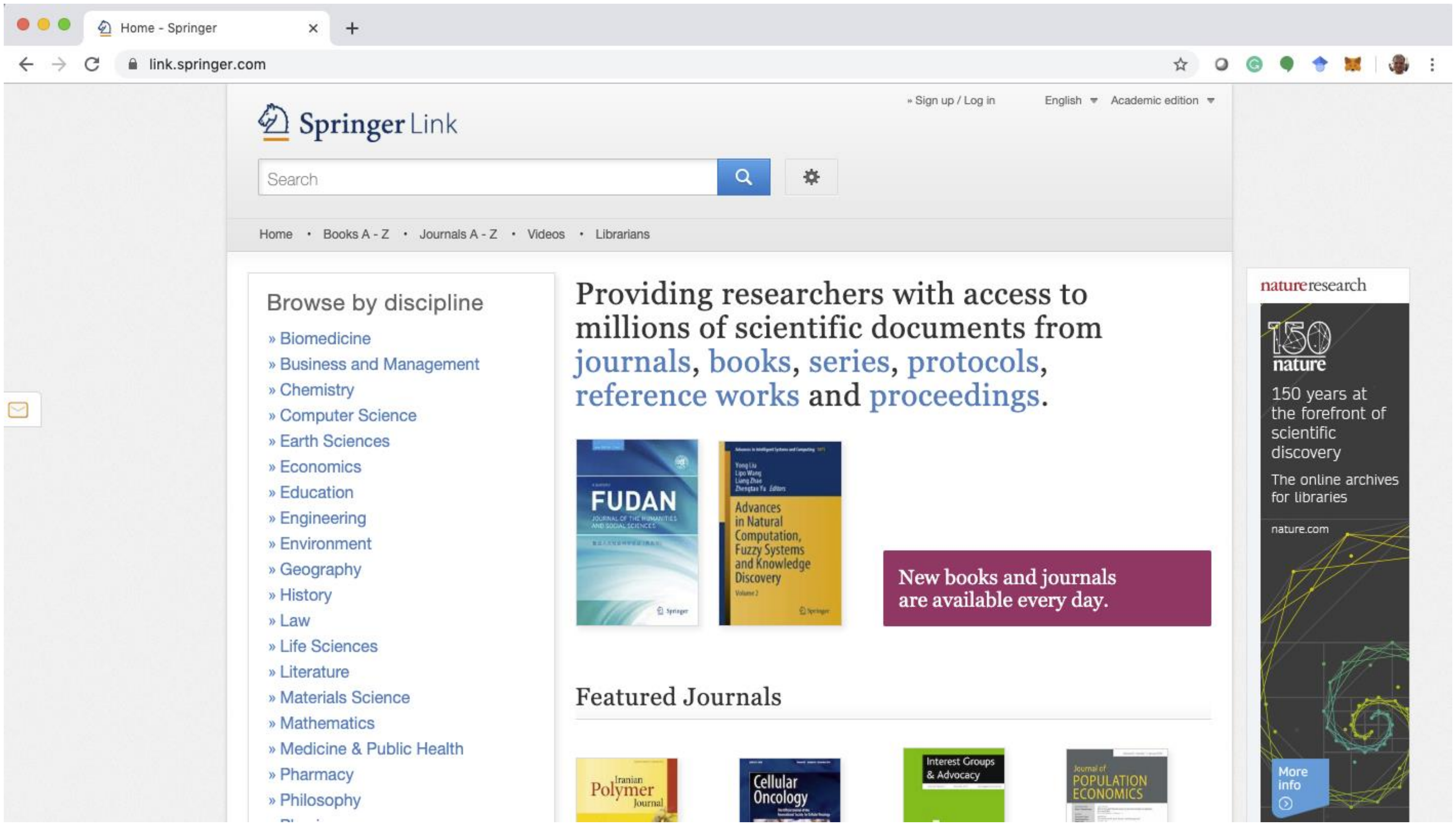
ACM Digital Library



ScienceDirect



SpringerLink



Accessing the resources

- Network connection at ULisboa
- Wireless network **eduroam**
- From outside:

  FCUL VPN (Setup for you)

Which papers to look for?

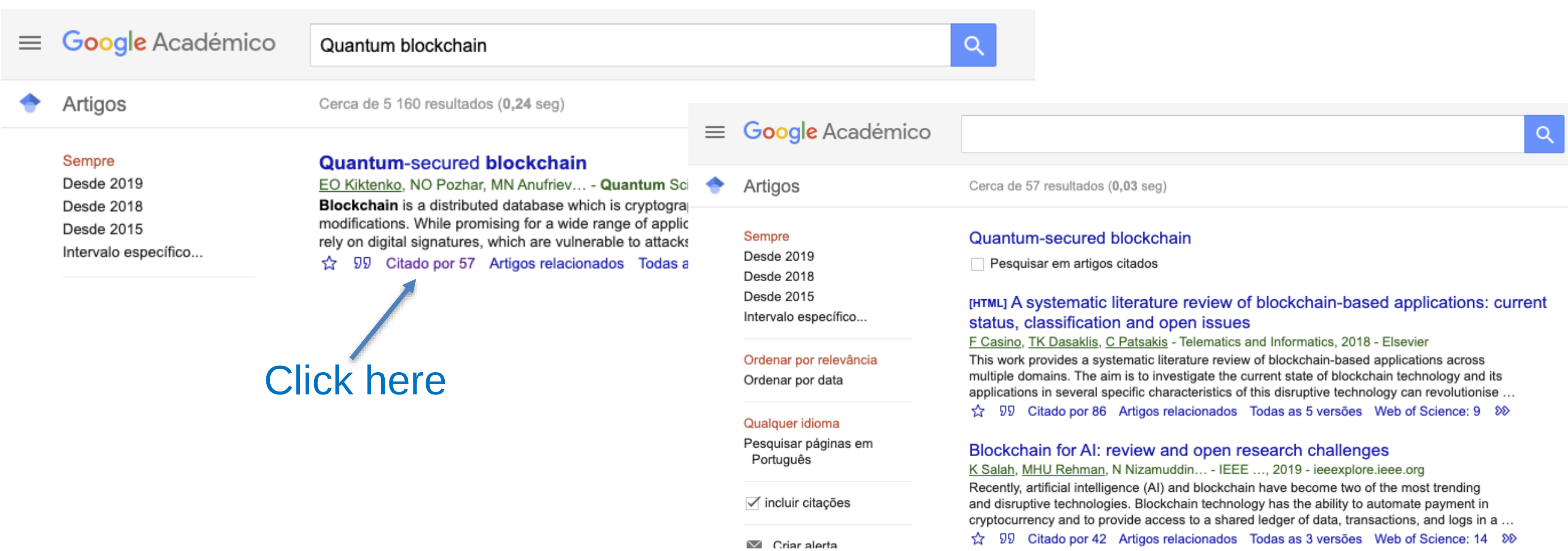
- Look for **keywords**
  - Be careful to not waste time reading low-quality or questionable papers
- Look for citations. **How many times** has that paper been cited? – The higher, the better
- Ask your **advisor and colleagues** for possible starting points and follow the lead
- Look for the best **conferences and journals** in your area and search for related work there (also good for knowing how to write)
- Follow the **key authors** in your field

Which papers to look for?

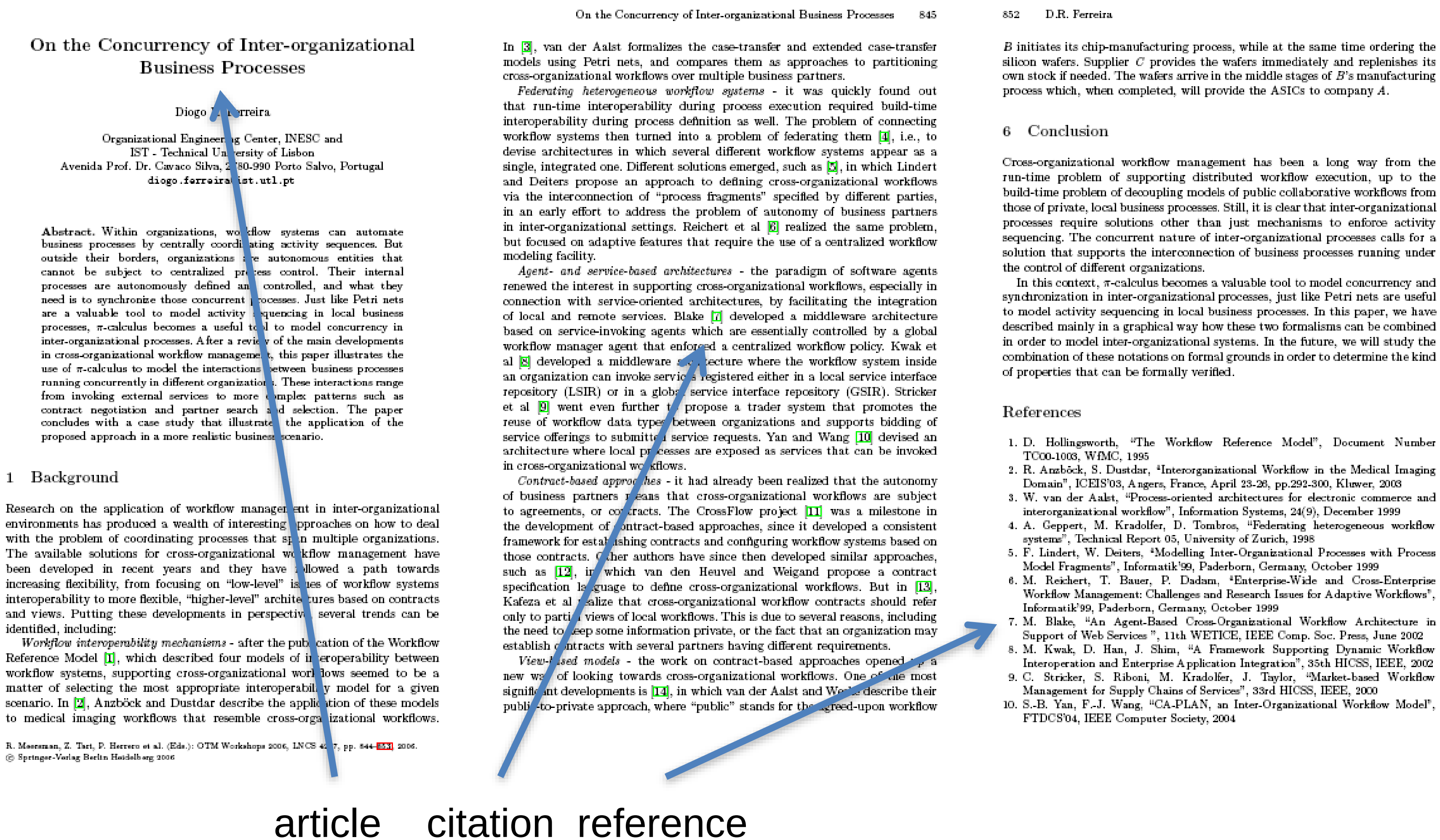
- Caveats:
  - (Again) Do not waste time with low-quality paper
    - Badly written or unpublished/published in a questionable venue
  - Even in the best venues there are some not so good papers (but you can learn something by trying to understand why they were accepted)
    - The opposite is also true!
  - Be open minded and diverse about the key authors
- Best thing is to combine all these approaches since they are complementary

Following the leads

- Enlarge your keyword set by reading the paper
- See the cited references: What happened before?
- See who cites it: What happened after?



Found an interesting article, now what?...



On the Concurrency of Inter-organizational Business Processes

Diogo R. Ferreira

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Abstract. Within organizations, workflow systems can automate business processes by centrally coordinating activity sequences. But outside their borders, organizations are autonomous entities that cannot be subject to centralized process control. Their internal processes are autonomously defined and controlled, and what they need is to synchronize those concurrent processes. Just like Petri nets are a valuable tool to model activity sequencing in local business processes, π-calculus becomes a useful tool to model concurrency in inter-organizational processes. After a review of the main developments in cross-organizational workflow management, this paper illustrates the use of π-calculus to model the interactions between business processes running concurrently in different organizations. These interactions range from invoking external services to more complex patterns such as contract negotiation and partner search and selection. The paper concludes with a case study that illustrates the application of the proposed approach in a more realistic business scenario.

1 Background

Research on the application of workflow management in inter-organizational environments has produced a wealth of interesting approaches on how to deal with the problem of coordinating processes that span multiple organizations. The available solutions for cross-organizational workflow management have been developed in recent years and they have followed a path towards increasing flexibility, from focusing on "low-level" issues of workflow systems interoperability to more flexible, "higher-level" architectures based on contracts and views. Putting these developments in perspective, several trends can be identified, including:

Workflow interoperability mechanisms - after the publication of the Workflow Reference Model [1], which described four models of interoperability between workflow systems, supporting cross-organizational workflows seemed to be a matter of selecting the most appropriate interoperability model for a given scenario. In [2], Anzböck and Dustdar describe the application of these models to medical imaging workflows that resemble cross-organizational workflows.

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In [3], van der Aalst formalizes the case-transfer and extended case-transfer models using Petri nets, and compares them as approaches to partitioning cross-organizational workflows over multiple business partners.

Federating heterogeneous workflow systems - it was quickly found out that run-time interoperability during process execution required build-time interoperability during process definition as well. The problem of connecting workflow systems then turned into a problem of federating them [4], i.e., to devise architectures in which several different workflow systems appear as a single, integrated one. Different solutions emerged, such as [5], in which Lindert and Deiters propose an approach to defining cross-organizational workflows via the interconnection of "process fragments" specified by different parties, in an early effort to address the problem of autonomy of business partners in inter-organizational settings. Reichert et al [6] realized the same problem, but focused on adaptive features that require the use of a centralized workflow modeling facility.

Agent- and service-based architectures - the paradigm of software agents renewed the interest in supporting cross-organizational workflows, especially in connection with service-oriented architectures, by facilitating the integration of local and remote services. Blake [7] developed a middleware architecture based on service-invoking agents which are essentially controlled by a global workflow manager agent that enforced a centralized workflow policy. Kwak et al [8] developed a middleware architecture where the workflow system inside an organization can invoke services registered either in a local service interface repository (LSIR) or in a global service interface repository (GSIR). Stricker et al [9] went even further to propose a trader system that promotes the reuse of workflow data types between organizations and supports bidding of service offerings to submitted service requests. Yan and Wang [10] devised an architecture where local processes are exposed as services that can be invoked in cross-organizational workflows.

Contract-based approaches - it had already been realized that the autonomy of business partners means that cross-organizational workflows are subject to agreements, or contracts. The CrossFlow project [11] was a milestone in the development of contract-based approaches, since it developed a consistent framework for establishing contracts and configuring workflow systems based on those contracts. Other authors have since then developed similar approaches, such as [12], in which van den Heuvel and Weigand propose a contract specification language to define cross-organizational workflows. But in [13], Kafeza et al realize that cross-organizational workflow contracts should refer only to partial views of local workflows. This is due to several reasons, including the need to keep some information private, or the fact that an organization may establish contracts with several partners having different requirements.

View-based models - the work on contract-based approaches opened up a new way of looking towards cross-organizational workflows. One of the most significant developments is [14], in which van der Aalst and Weske describe their public-to-private approach, where "public" stands for the agreed-upon workflow

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B initiates its chip-manufacturing process, while at the same time ordering the silicon wafers. Supplier C provides the wafers immediately and replenishes its own stock if needed. The wafers arrive in the middle stages of B's manufacturing process which, when completed, will provide the ASICs to company A.

6 Conclusion

Cross-organizational workflow management has been a long way from the run-time problem of supporting distributed workflow execution, up to the build-time problem of decoupling models of public collaborative workflows from those of private, local business processes. Still, it is clear that inter-organizational processes require solutions other than just mechanisms to enforce activity sequencing. The concurrent nature of inter-organizational processes calls for a solution that supports the interconnection of business processes running under the control of different organizations.

In this context, π-calculus becomes a valuable tool to model concurrency and synchronization in inter-organizational processes, just like Petri nets are useful to model activity sequencing in local business processes. In this paper, we have described mainly in a graphical way how these two formalisms can be combined in order to model inter-organizational systems. In the future, we will study the combination of these notations on formal grounds in order to determine the kind of properties that can be formally verified.

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BibTeX entry types

- Book
- Book chapter (*incollection*)
- Published paper
  - In a scientific journal (*article*)
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- …
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Book

- Springer
  - Foster, I., Kesselman, C.: The Grid: Blueprint for a New Computing Infrastructure. Morgan Kaufmann, San Francisco (1999)
- MLA (*Modern Languages Association, common in the humanities*)
  - Hodgkinson, Tom. *How to Be Idle*. New York: Harper, 2005.
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Citations

- Numeric style associated to citations

  Some techniques use probabilistic models [17,18] but most of the current techniques are geared towards retrieving Petri net models [19].

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Citations

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  - One author
    - A recent study found a possible genetic cause of alcoholism (Pauling, 2005). Pauling (2005) discovered a possible genetic cause of alcoholism.
  - Two authors
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    - A recent study found a possible genetic cause of alcoholism (Pauling et al., 2005). Pauling et al. (2005) discovered a possible genetic cause of alcoholism.
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    - Recent studies found a possible genetic cause of alcoholism (Alford, 1995; Pauling et al., 2005; Sirkis, 2003).

Citations

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    title = "{On Computable Numbers, with an Application to the
              Entscheidungsproblem}",
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}
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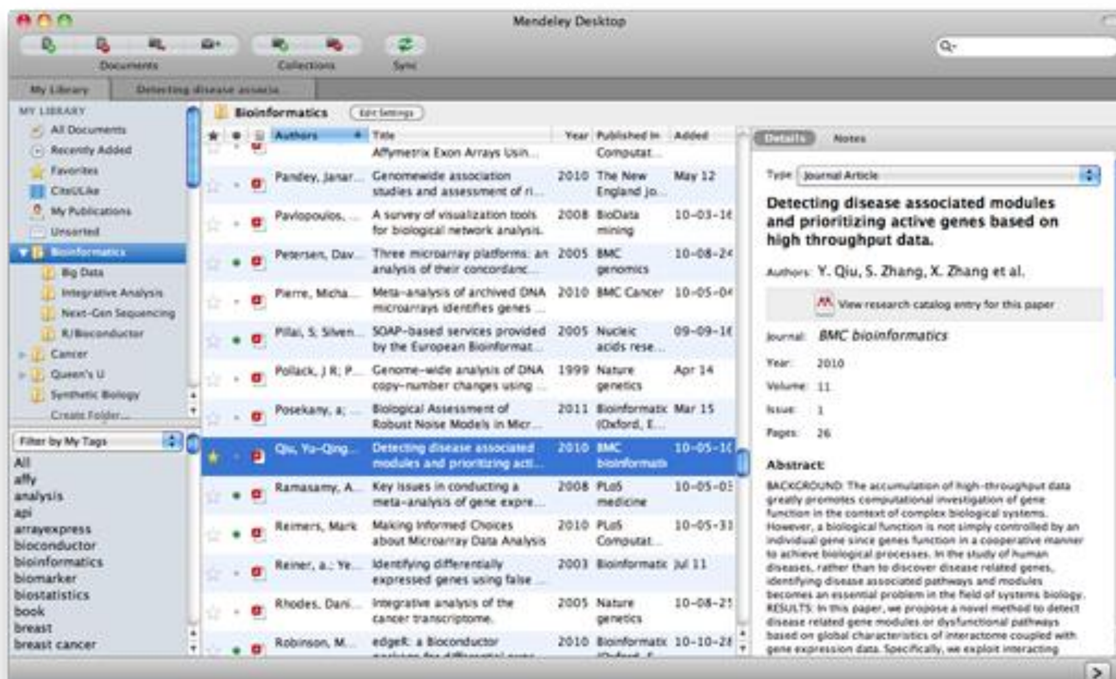

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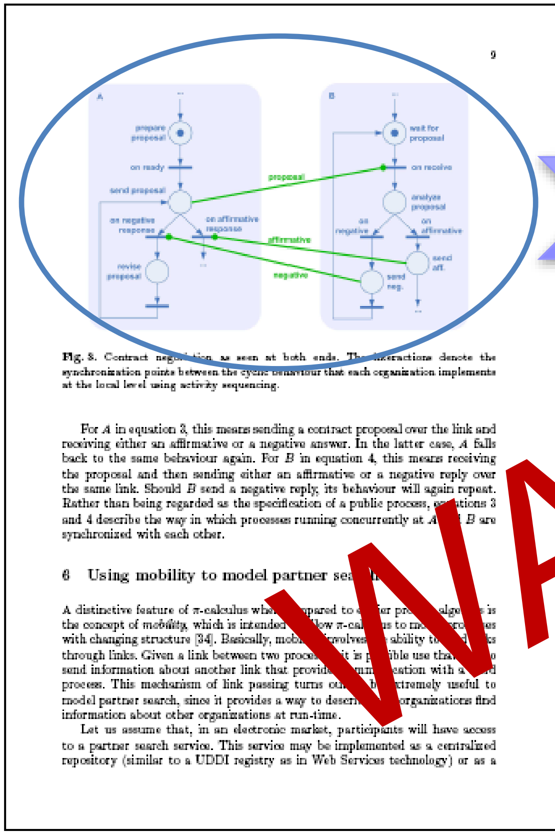

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Fig. 8. Contract neg[illegible] as seen at both ends. Th[illegible]nteractions denote the synchronization points between the cyclic [illegible] that each organization implements at the local level using activity sequencing.

For A in equation 3, this means sending a contract proposal over the link and receiving either an affirmative or a negative answer. In the latter case, A falls back to the same behaviour again. For B in equation 4, this means receiving the proposal and then sending either an affirmative or a negative reply over the same link. Should B send a negative reply, its behaviour will again repeat. Rather than being regarded as the specification of a public process, e[illegible]ations 3 and 4 describe the way in which processes running concurrently at A [illegible] B are synchronized with each other.

6 Using mobility to model partner se[illegible]

A distinctive feature of π-calculus whe[illegible]pared to e[illegible]er pro[illegible] alge[illegible] is the concept of *mobility*, which is intended [illegible]low π-cal[illegible]us to mo[illegible]ro[illegible]es with changing structure [34]. Basically, mob[illegible] involves [illegible] ability t[illegible] [illegible]ks through links. Given a link between two proce[illegible] it is p[illegible]ible use tha[illegible] send information about another link that provide[illegible] [illegible]ation with a [illegible]d process. This mechanism of link passing turns ou[illegible] b[illegible] extremely useful to model partner search, since it provides a way to descri[illegible] organizations find information about other organizations at run-time.

Let us assume that, in an electronic market, participants will have access to a partner search service. This service may be implemented as a centralized repository (similar to a UDDI registry as in Web Services technology) or as a

Some previous article

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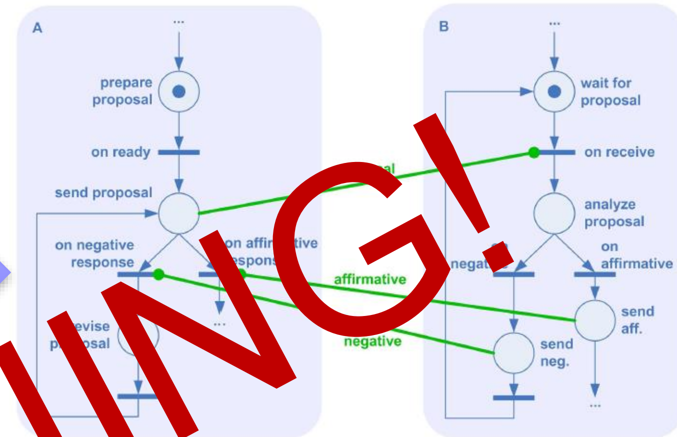


Figure 1 – An interesting diagram.

Figure 1 presents an interesting diagram, combining the notations of Petri Nets and Pi calculus. The first has been used to model (…)

My own work

Found an interesting figure...

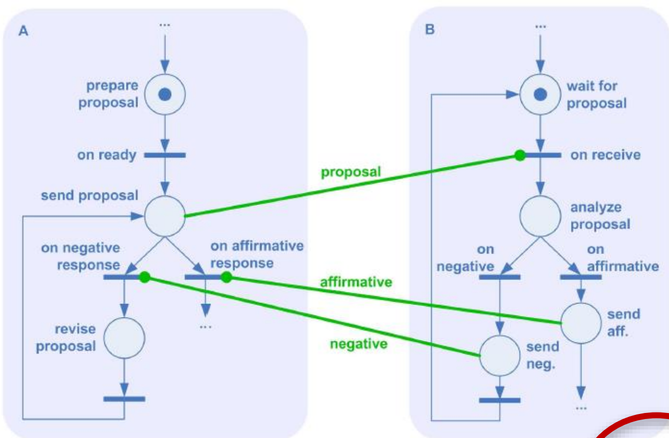


Figure 1. Interesting diagram [20]

Figure 1 presents an interesting diagram, combining the notations of Petri Nets and Pi calculus. The first has been used to model (…)

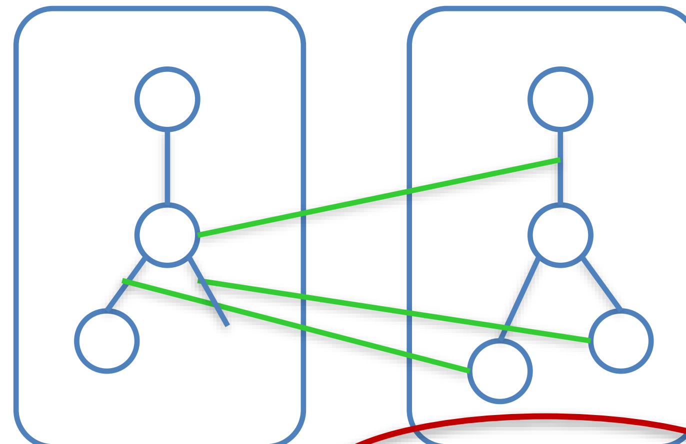
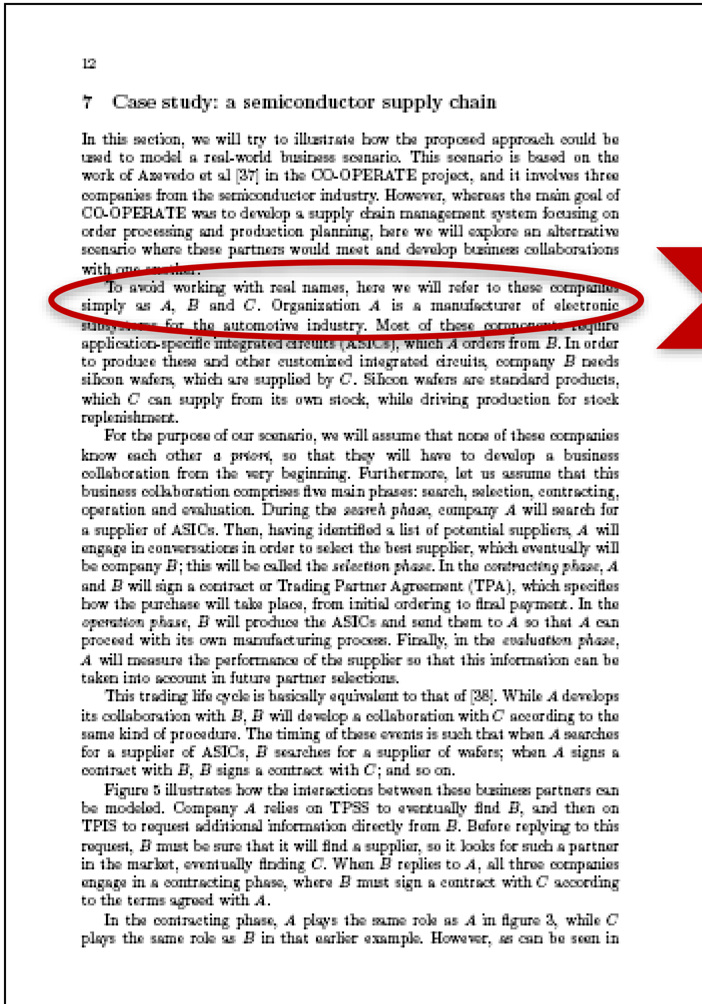


Figure 1. Diagram (adapted from [20])

Figure 1 presents an interesting diagram, combining the notations of Petri Nets and Pi calculus. The first has been used to model (…)

(in any case, do not exagerate on the presentation of such figures)

Found an interesting statement...



12

7 Case study: a semiconductor supply chain

In this section, we will try to illustrate how the proposed approach could be used to model a real-world business scenario. This scenario is based on the work of Azevedo et al [37] in the CO-OPERATE project, and it involves three companies from the semiconductor industry. However, whereas the main goal of CO-OPERATE was to develop a supply chain management system focusing on order processing and production planning, here we will explore an alternative scenario where these partners would meet and develop business collaborations with one another.

To avoid working with real names, here we will refer to these companies simply as A, B and C. Organization A is a manufacturer of electronic subsystems for the automotive industry. Most of these components require application-specific integrated circuits (ASICs), which A orders from B. In order to produce these and other customized integrated circuits, company B needs silicon wafers, which are supplied by C. Silicon wafers are standard products, which C can supply from its own stock, while driving production for stock replenishment.

For the purpose of our scenario, we will assume that none of these companies know each other *a priori*, so that they will have to develop a business collaboration from the very beginning. Furthermore, let us assume that this business collaboration comprises five main phases: search, selection, contracting, operation and evaluation. During the *search phase*, company A will search for a supplier of ASICs. Then, having identified a list of potential suppliers, A will engage in conversations in order to select the best supplier, which eventually will be company B; this will be called the *selection phase*. In the *contracting phase*, A and B will sign a contract or Trading Partner Agreement (TPA), which specifies how the purchase will take place, from initial ordering to final payment. In the *operation phase*, B will produce the ASICs and send them to A so that A can proceed with its own manufacturing process. Finally, in the *evaluation phase*, A will measure the performance of the supplier so that this information can be taken into account in future partner selections.

This trading life cycle is basically equivalent to that of [38]. While A develops its collaboration with B, B will develop a collaboration with C according to the same kind of procedure. The timing of these events is such that when A searches for a supplier of ASICs, B searches for a supplier of wafers; when A signs a contract with B, B signs a contract with C; and so on.

Figure 5 illustrates how the interactions between these business partners can be modeled. Company A relies on TPSS to eventually find B, and then on TPIS to request additional information directly from B. Before replying to this request, B must be sure that it will find a supplier, so it looks for such a partner in the market, eventually finding C. When B replies to A, all three companies engage in a contracting phase, where B must sign a contract with C according to the terms agreed with A.

In the contracting phase, A plays the same role as A in figure 3, while C plays the same role as B in that earlier example. However, as can be seen in

Interesting statement, that I should use in my article…

Monopoly

Jail Bait (1937)

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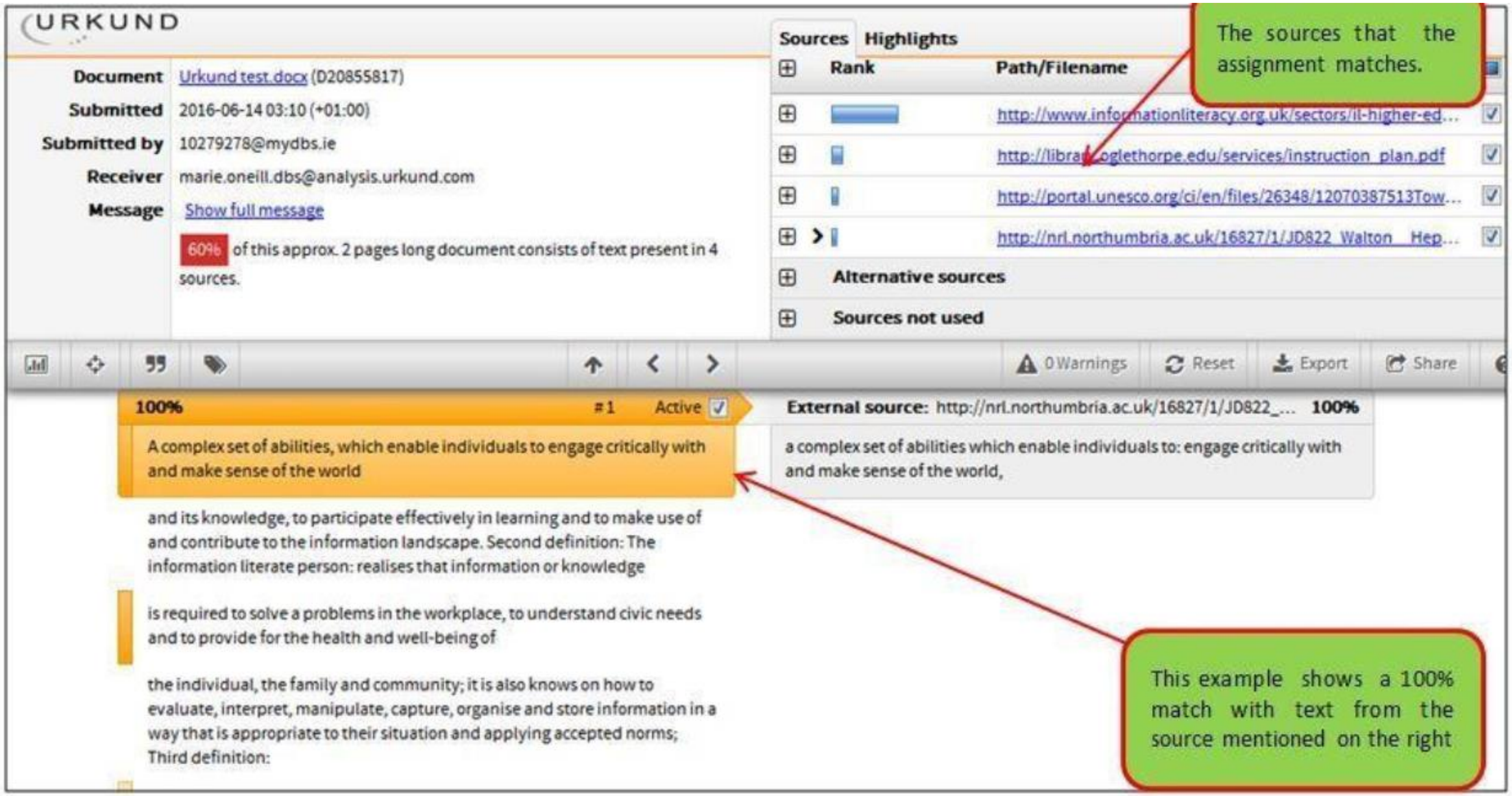

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[1]

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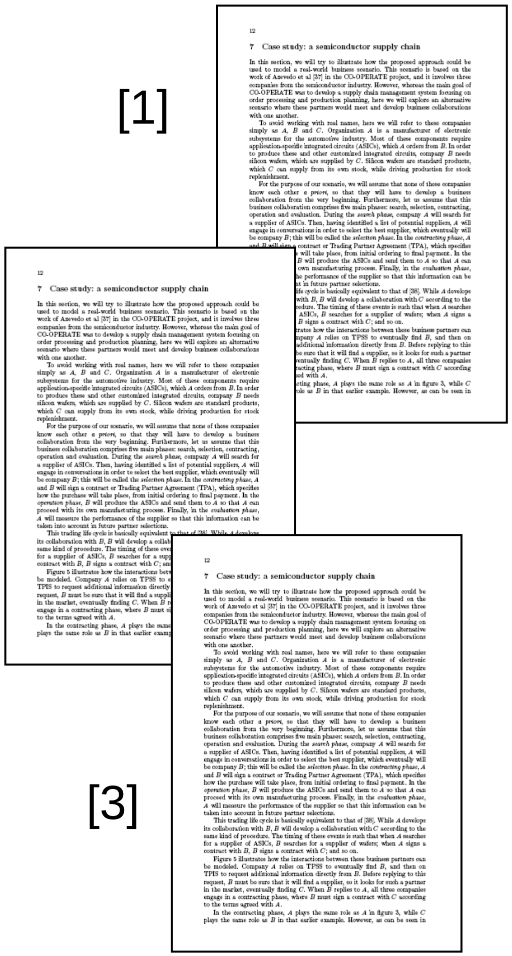

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For the purpose of our scenario, we will assume that none of these companies know each other *a priori*, so that they will have to develop a business collaboration from the very beginning. Furthermore, let us assume that this business collaboration comprises five main phases: search, selection, contracting, operation and evaluation. During the *search phase*, company A will search for a supplier of ASICs. Then, having identified a list of potential suppliers, A will engage in conversations in order to select the best supplier, which eventually will be company B; this will be called the *selection phase*. In the *contracting phase*, A and B will sign a contract or Trading Partner Agreement (TPA), which specifies how the purchase will take place, from initial ordering to final payment. In the *operation phase*, B will produce the ASICs and send them to A so that A can proceed with its own manufacturing process. Finally, in the *evaluation phase*, A will measure the performance of the supplier so that this information can be taken into account in future partner selections.

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…

Some previous authors have argued for the first theory [1,2]. Still, others have supported a second different theory [3].

In this work, we have developed an alternative theory (…)

…

Autor desconhecido

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- One of the best things an MSc student can achieve is to be a author or co author on a scientific paper during his/her MSc –
  - **This is the "Holy Grail" of a MSc work!**
  - Bibliographic research is **absolutely critical for a decent paper**
  - This will have a direct impact on the final grade
  - Will dramatically increase the chances of getting to a PhD
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  - First Author in Peer Reviewed international journal
  - First Author in oral presentation in Core A* Conference
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Estudo Orientado em Engenharia Informática 2022/2023

Making presentations

Andre Falcao (aofalcao@fc.ul.pt)

Parts of this presentation were originally prepared by Prof. Bruno Martins (IST) and Alysson Bessani (FCUL)

Summary

- What is a good presentation?
- Presentation tips and pitfalls
  - Presentation organization
  - Making slides
  - Using charts
- Factors for an engaging presentation

Presenting papers at conferences

Understanding Spaghetti Models with
Sequence Clustering for ProM

Gabriel M. Veiga and Diogo R. Ferreira

IST – Technical University of Lisbon
Avenida Prof. Dr. Cavaco Silva
2744-016 Porto Salvo, Portugal
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Abstract. The goal of process mining is to discover process models from event logs. However, for processes that are not well structured and have a lot of diverse behavior, existing process mining techniques generate highly complex models that are often difficult to understand; these are called spaghetti models. One way to try to understand these models is to divide the log into clusters in order to analyze reduced sets of cases. However, the amount of noise and ad-hoc behavior present in real-world logs still poses a problem, as this type of behavior interferes with the clustering and complicates the models of the generated clusters, affecting the discovery of patterns. In this paper we present an approach that aims at overcoming these difficulties by extracting only the useful data and presenting it in an understandable manner. The solution has been implemented in ProM and is divided in two stages: preprocessing and sequence clustering. We illustrate the approach in a case study where it becomes possible to identify behavioral patterns even in the presence of very diverse and confusing behavior.

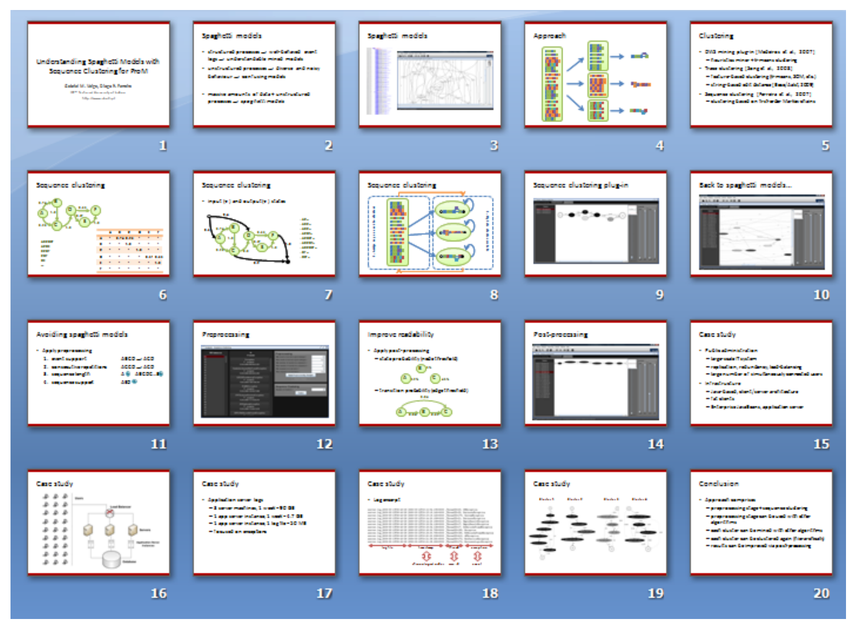
Key words: Process Mining, Preprocessing, Sequence Clustering, ProM, Markov Chains, Event Logs, Hierarchical Clustering, Process Models

1 Introduction

The main application of process mining is the discovery of process models. For processes with a lot of different cases and high diversity of behavior, the models generated tend to be very confusing and difficult to understand. These models are usually called *spaghetti models*. Clustering techniques have been investigated as a means to deal with this complexity by dividing cases into clusters, leading to less confusing models. However, results may still suffer from the presence of certain unusual cases that include noise and ad-hoc behavior, which are common in real-world environments. Usually this type of behavior is not relevant to understand a process and it unnecessarily complicates the discovered models.

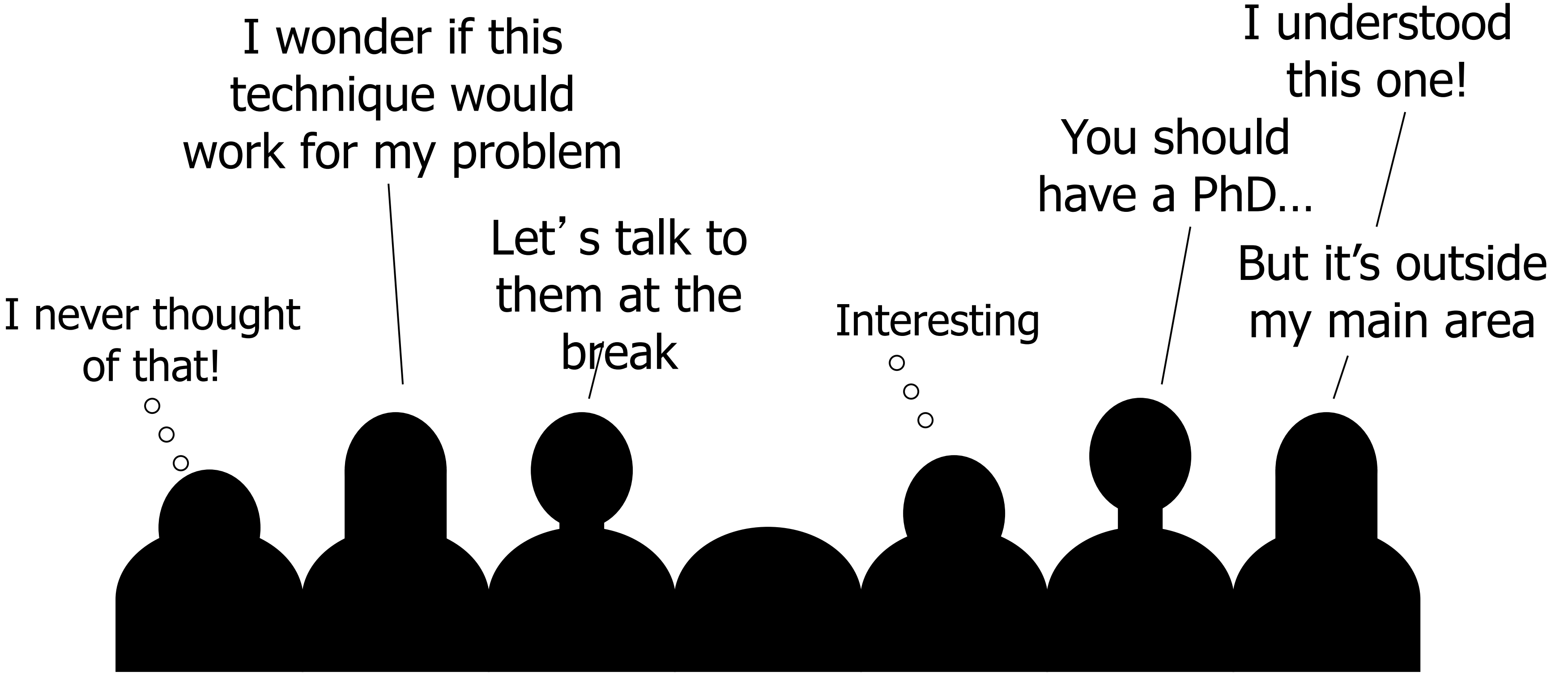
In this paper we present an approach that is able to deal with these problems by means of sequence clustering techniques. This is a kind of model-based clustering that partitions the cases according to the order in which events occurred. For the purpose of this work the model used to represent each cluster is

- **Typical Presentation**
  - 15-20min + 5 min Q&A
  - Point to a max. 20 slides



Good presentations

- Give the audience a sense of what your work is…
- Make them want to read your paper…



General tips for presenting

- Start by determining the main ideas you want to convey
  - **Spoil the punch line, by stating your results early and in simple terms**
- Time is generally short, so you must be efficient
- Research your audience's background and target the presentation to them
- Speak clearly and directly
  - Think of the presentation as a performance
  - Articulate clearly
  - Limit jokes, side comments, "*ums*", and "*ahs*", etc.
  - Look at the audience
- **Rehearse!**

Organize your presentation

- Begin with an outline <- Only in seminars
- **Provide motivation for the topic**
- Summarize your approach
  - Limit Maths! in most talks no one is going to read complex equations
  - Contrast your approach against previous work
    - **Why is your approach needed (it's not necessarily better)**
- Present results
- Discuss results and their implications
- Summarize
- Prepare back-up slides to answer questions

Slides = Visual Aids

- Minimize text
  - Keep less than 8 bullets per page
  - Do not use long sentences
- Keep text legible
  - Using font size larger than 20pt is desirable
  - Dark text on light background or reverse
- **Practical advice** : *Limit animations!*
- Use tables sparingly and keep them simple (*and charts are much better here*)
  - tables can appear in the main document

Never forget

- The presentation reminds the audience of the **most important findings and their implications**…
- Detailed information is on the Main Document (Thesis, Report or Paper)!
- Corollary: Do not put everything in the presentation, only the relevant stuff

Slide layout – the bad example

- This page contains too many words for a presentation slide. It is not written in point form, making it difficult both for your audience to read and for you to present each point. Although there are exactly the same number of points on this slide as the next slide, it looks much more complicated. In short, your audience will spend too much time trying to read this paragraph instead of listening to you.

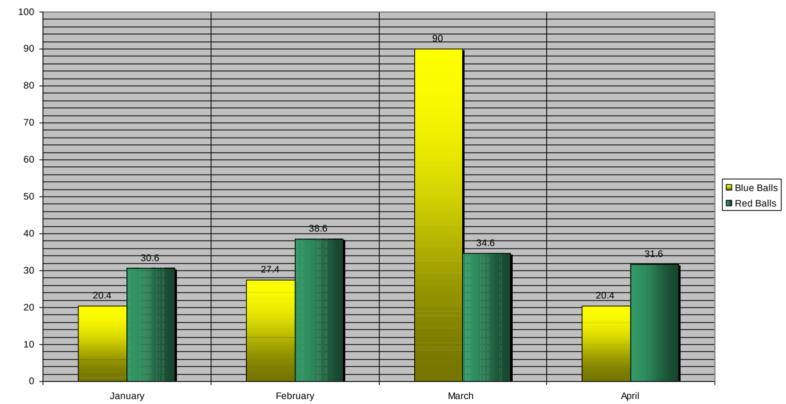
Slide layout – the good example

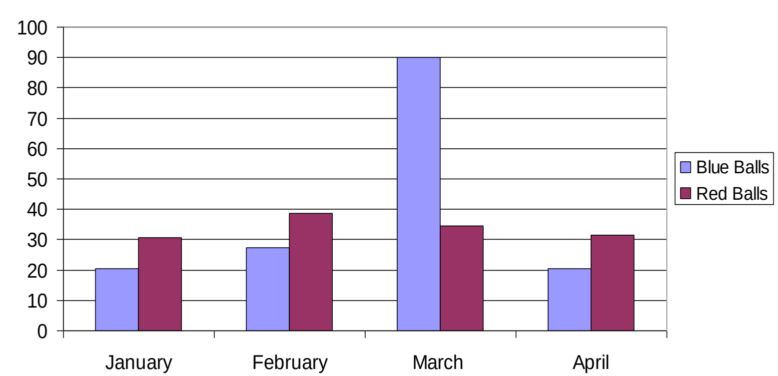
- Show one point at a time:
  - Will help audience concentrate on what you are saying
  - Will prevent audience from reading ahead
  - Will help you keep your presentation focused
- Slides support, but do not replace, your discussion
- **Do not read directly from the slides or your notes**

Figures and charts

- Some guidelines for figures
  - Simpler is better in presentations
  - Keep text legible
  - Choose strongly contrasting colors
  - Include error/data info
  - Explain your figures
  - Shading is distracting
  - Minor gridlines are unnecessary
  - Don't use 3D in columns or bars
  - Avoid mind numbing Excel charts

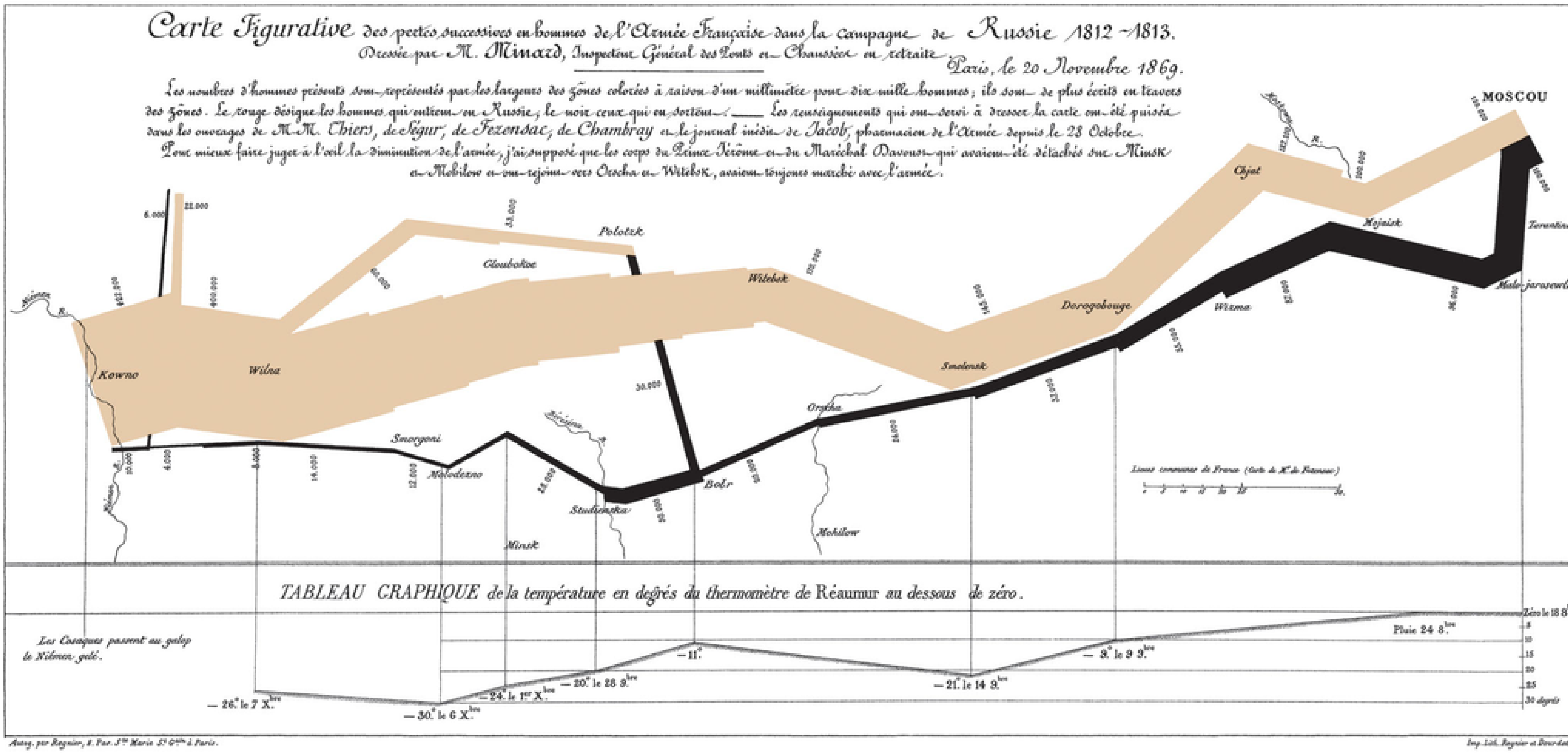
- Limit the number of figures





The perfect example

(by Charles Minard)



Time allotted for presentation and questions

Rookie's Notes:

- Allow for approx. 1-1.5 minutes per slide
  - In good, really good presentations this rule-of-thumb can be VASTLY different
- Allocate time in proportion to the topic's importance
- Rehearse at least twice for timing
- Simplify or remove slides, to adjust timing
- Sleep on it!

The most important thing

- **Tell a story and believe in it!**
  - If you do not believe what you are presenting the others also won't
  - Your presentation should be pleasant
  - Show enthusiasm but not sillyness
  - No breaks. Everything should flow like it was the absolute logical consequence of the previous events

Last, but not the least

- Do not even think of presenting something you do not understand
  - It will bite you and EVERYONE will notice
- Invaluable tip: Use the **Feynman Technique**
  1. Learn a concept that you need
  2. Pretend you are explaining it to a young student, in your own words
  3. Identify what you cannot explain well and revise and repeat previous steps until it is perfect
  4. Eventually review and simplify

Questions?